

School of GeoSciences

Dissertation
for the degree of

MSc in MARINE SYSTEMS AND POLICIES

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August 2025

Nature-based Solutions to Estuarine Eutrophication:

The effectiveness of multi-cell constructed wetlands in mitigating nutrient enrichment at Budle Bay, Lindisfarne.



Source: LIFE WADER (2025c)

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In collaboration with: LIFE WADER, & Natural England.

Abstract

Eutrophication in estuarine environments, driven by nutrient-rich agricultural runoff, causes ecological degradation, including harmful algal blooms and biodiversity loss. Constructed wetlands (CWs) have emerged as nature-based solutions for attenuating nutrient pollution through physical, chemical and biological processes, improving downstream water quality. Agricultural runoff has caused eutrophication and harmful algal blooms in the Lindisfarne National Nature Reserve (LNNR), particularly at Budle Bay. This study investigates the effectiveness of a pilot multi-cell CW at Cockle Ridge, one of two priority intervention sites identified by Natural England within Budle Bay, with potential application at the second site, Kiln Point.

Water and sediment samples were collected in May 2025 from six locations across both sites, alongside post-construction observations. Samples were analysed for total oxidised nitrogen (TON), phosphorus, and ammonium concentrations using spectrophotometry. Spatial nutrient and vegetation variations within the pilot intervention were assessed to determine treatment effectiveness across the four wetland cells and identify operating wetland processes. Kiln Point nutrient concentrations were compared against Cockle Ridge to predict treatment of the proposed 5-cell CW. Historical Environment Agency and Natural England datasets for Cockle Ridge provided pre-construction baselines for comparison, with only limited data available for Kiln Point.

Results demonstrated substantial TON reductions at Cockle Ridge compared to pre-intervention levels (95%) and across the intervention (96.4%), with the first cell providing most treatment (77.5%). These results are particularly noteworthy given the TON-dominated catchment. Ammonium concentrations increased (88.3%), while phosphorus showed modest reduction (11.6%) compared to pre-intervention levels, indicating needs for continued monitoring and additional nature-based strategies. This result justifies implementing CWs at Kiln Point, another TON dominated catchment. Extrapolated results suggest the proposed 5-cell CW could achieve a 96.59% TON reduction. Additional baseline data for Kiln Point is required before CW implementation. This study supports wider CW implementation as effective nature-based solutions for improving estuarine water quality in TON-dominated catchments. Recommendations include developing a standardised monitoring methodology and prioritising single-cell CWs when constrained by budget or land availability given their substantial nutrient removal contribution.

(Word Count: 14,970)

Acknowledgements

I would like to thank Dr Sebastian Hennige for his continuous support throughout this process, for answering endless questions and for always providing a positive outlook for the many challenges I came across. I extend my thanks to all those at Natural England who have helped me through fieldwork, project questions, and providing datasets, particularly Amelia Welch and Mark Renfrey, as well as Anthony Crook from the Environment Agency. I'm also grateful to LIFE WADER and the Berwickshire and Northumberland Marine Nature Partnership for enabling this dissertation.

I am eternally grateful to Gavin Sim and Yang Bai for all their help in the hours of lab work undertaken for this project. Thank you for putting up with continuous requests for more mixed reagent and centrifuge tubes. Special thanks also go to Maz Bui, an amazing friend and fieldwork partner. The hours spent syringing water samples and the permanent indentation left in your hand as a result will forever be remembered. This project wouldn't have been possible without you. I also want to acknowledge my trusty car, by no means a worthy fieldwork vehicle, but which showed its undeniable potential during the return journey from our fieldwork.

Thanks must also go to my Mum and Amy Stocks for spending hours reading over my many drafts, your advice was invaluable. Finally, thank you to my friends and family for providing much needed distractions throughout this lengthy dissertation project, from quests for sweet treats and home-cooked meals, to listening to my many rants.

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Equation 1: $\text{mg L}^{-1} = \mu mol\ \text{L}^{-1} \times (Molecular\ Weight \div 1000)$

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$$Final\ Water\ Concentration\ (\text{mg L}^{-1}) = UV - Vis\ Concentration\ (\text{mg L}^{-1}) \times Dilution\ Factor$$

Equation 3:

$$Final\ Sediment\ Concentration\ (\mathrm{mg}\ kg^{-1}) = \frac{UV\text{–}Vis\ Concentration\ (\mathrm{mg\ L^{-1}}) \times Dilution\ Factor \times Extraction\ Volume\ (ml)}{\frac{Dry\ Mass\ (g)}{Wet\ Mass\ (g)} \times Extraction\ Mass}$$

List of Abbreviations

CW	Constructed Wetland
DNRA	Dissimilatory Nitrate Reduction to Ammonium
EA	Environment Agency
EU	European Union
HAB	Harmful Algal Bloom
HRT	Hydraulic Residence Time
IQR	Inter-quartile Range
LNNR	Lindisfarne National Nature Reserve
MSFD	Marine Strategy Framework Directive
NbS	Nature-based Solution
NE	Natural England
NVZ	Nitrate Vulnerable Zone
RPM	Revolutions per Minute
SAC	Special Area of Conservation
SD	Standard deviation
SFCW	Surface Flow Constructed Wetland
SPA	Special Protection Area
SSF	Subsurface Flow
SSSI	Special Site of Scientific Interest
TON	Total Oxidised Nitrogen
WADER	Water and Disturbance Environmental Restoration
WFD	Water Framework Directive
WPA	Water Priority Area
WwTW	Wastewater Treatment Work
UK	United Kingdom

1 Introduction

Scientific Context

1.1 Global Issues of Estuary Pollution

Estuaries are dynamic semi-enclosed coastal waterbodies where freshwater meets the open sea (Cardoso, 2022). Despite comprising only 7% of marine environments, these highly productive habitats provide over 50% of food for ocean ecosystems due to being rich in nitrogen and phosphorus (Alexander and Fairbridge, 1999; Cardoso, 2022; Häder *et al.,* 2020). With over 2.15 billion people living in near-coastal zones, estuarine ecosystems face mounting anthropogenic stressors including nutrient enrichment, habitat loss, increased tourism, invasive species, and industrial development (Reimann *et al.,* 2023; UN, 2016). Approximately, 80% of marine pollution globally originates from river and surface runoff inputs, making diffuse pollution the primary threat to estuarine water quality (Mallin *et al.,* 2012; UN, 2016). Runoff represents one of the most persistent and difficult pollution management challenges (Matthews *et al*., 2024; Turcios *et al*., 2021).

Growing interest in Nature-based Solutions (NbS) over the past decade has emerged as a promising approach for ecosystem management (Cassin, 2021). These solutions sustainably address environmental challenges through natural processes while delivering additional benefits for human well-being and biodiversity, offering cost-effective alternatives to energy-intensive engineered infrastructure (Barman *et al*., 2024; Cohen-Shacham *et al.,* 2016; Nesshöver *et al.,* 2017).

1.2 Nutrient Pollution

In temperate estuarine ecosystems, runoff and discharges rich in ammonium, nitrate, and nitrite are the main cause of ecosystem damage (Deng *et al.,* 2015; Yousaf *et al.,* 2021). Hereafter, nitrate and nitrite are combined and referred to as the total oxidised nitrogen (TON). These nutrients greatly impact aquatic systems as they are readily available for uptake by plants and microorganisms in contrast to other particulate organic nitrogenous species like amino acids and urea (Jamieson *et al.,* 2003; Yousaf *et al.,* 2021). In the tropics, phosphorus is thought to be more limiting on primary productivity than nitrogen, but it can also contribute to temperate coastal ecosystem damage (Cabral *et al.,* 2019; Howarth, 2008).

1.3 Impacts of Nutrient Pollution on Estuaries

Eutrophication has long been understood to result from excessive nitrogen and phosphorus enrichment (Häder *et al.,* 2021; Howarth 1996). This causes harmful algal blooms (HABs), phytoplankton and cyanobacteria growth, and dissolved oxygen reduction as algae decompose (Jamieson *et al.,* 2003; Mallin *et al.,* 2012). These hypoxic conditions degrade water quality, reduce biodiversity, and frequently cause fish mortality, representing one of the greatest threats to marine ecosystem health (Dorgham, 2014; Devlin and Brodie, 2023; Nadarajan and Sukumaran, 2021; Weeber *et al.,* 2023).

Mass occurrences of HABs have increased globally over recent decades, a phenomenon known as 'green tides' in estuarine environments caused by extensive reproduction of species from the *Ulva* genus (*Figure 1.1*) (Gubelit, 2022; Häder *et al.,* 2021; Kudela *et al.,* 2010; Smetacek and Zingone, 2013). Growing evidence links this increase in HAB frequency and magnitude, to enhanced nutrient availability (Kudela *et al.,* 2010).



Figure 1.1: Harmful algal bloom (HAB) visible from space. Recorded in the Douarnenez Bay, Brittany, France in June 2023. Sourced from European Space Agency (2023a).

Recognising this connection, the EU Water Framework Directive (WFD) (Directive 2000/60/EC) and Marine Strategy Framework Directive (MSFD) (Directive 2008/56/EC) classify HABs as biological quality elements, enabling their use as environmental

assessment indicators of eutrophication (European Commission, 2000; European Commission, 2008; Gubelit, 2022). Recent research has increasingly focused on implementing NbSs in estuarine environments to combat eutrophication by reducing nutrient-rich runoff inputs (van Rees *et al.,* 2023).

1.4 Agricultural Pollution

As the agriculture industry expands to meet growing food demands, organic and inorganic fertilisers are used in excess which, after heavy rainfall, leads to nutrient-rich surface runoff transported by rivers into the wider ecosystem, ultimately reaching lakes and estuaries (DEFRA, 2023; Deng *et al.,* 2024; Turcios *et al.,* 2021). The diffuse nature of this pollution makes it challenging to manage (Nsenga Kumwimba *et al.,* 2023).

In many European countries, waterbodies are failing to meet the EU WFD objective of 'good ecological status' (European Commission, 2000; Collins and Anthony, 2008; Holden *et al.,* 2017). This results in elevated nutrient concentrations within coastal waters causing eutrophication problems and impacting ecological processes (Cardoso, 2022; Environment Agency, 2007; Kennish and de Jonge, 2011). Widespread eutrophication demonstrates the urgent need for better agricultural runoff management (Abonyi *et al.,* 2024; Weeber *et al.,* 2023).

Groundwater represents another substantial nutrient source to estuaries beyond surface runoff (Wilson *et al.,* 2024). While groundwater can be naturally nutrient-rich due to hydrogeological conditions, anthropogenic activities including agriculture and industry substantially elevate nutrient concentrations in coastal watersheds (Robinson, 2015; Valiela, *et al.,* 1990). This contaminated groundwater subsequently causes eutrophication in discharge areas, creating another form of diffuse pollution that compounds the management challenges already described (Pieterse *et al.,* 2005).

1.5 Agricultural Pollution in the UK

In the UK, agriculture is responsible for 60% of TON and 25% of phosphorus in water bodies (Holden *et al.,* 2017; White and Hammond*,* 2009). Only 24% of water bodies in England, 36% in Wales, 65% in Scotland, and 22% in Northern Ireland meet the EU WFD 'good ecological status' (Holden *et al.,* 2017). Therefore, in 2023, the UK government proposed the Environmental Targets (Water)(England) Regulations in which agriculture targets aim to reduce nitrogen, phosphorus and sediment pollution into the water environment by at least

40% compared to the 2018 baseline, by 2038 (DEFRA, 2023). In the UK, agricultural pollution, causing water quality issues, costs approximately £234 million annually in associated treatment costs (Jacobs UK, 2008).

A notable case study of these national pollution patterns is found at the protected Lindisfarne National Nature Reserve (LNNR), where agricultural runoff has created substantial eutrophication challenges (Jarvie *et al.,* 2002). Budle Bay, with its predominantly agricultural catchment and intensive farming practices, experiences significantly greater nutrient loading than Holy Island Bay, while its shallow, sheltered topography and multiple field drain inputs create particularly vulnerable conditions for nutrient accumulation (Palmer, 2012; LIFE WADER, 2025a).

1.6 Treatment Approaches

Traditional wastewater treatment methods, such as sewage pumping stations and wastewater treatment works (WwTWs), cannot treat diffuse agricultural pollution which enters estuaries through dispersed surface runoff and small streams (Bi *et al*., 2024; Fan *et al.,* 2024; Ferreira *et al.,* 2023; Nsenga Kumwimba *et al*., 2023).

To address this challenge, agricultural pollution can be managed through good agricultural practices such as avoiding fertiliser application before predicted heavy rainfall and strategic ploughing to increase soil infiltration and reduce soil erosion (Haygarth *et al.,* 2009; Holden *et al.,* 2017; Preedy *et al.,* 2001). Additionally, constructed wetlands and buffer strips are common interventions of runoff that slow water and remove entrained nutrients through biological processes before they reach downstream receiving estuaries (Holden *et al.,* 2017; Ockenden *et al.,* 2012; Xia *et al.,* 2020).

Among these interventions, constructed wetlands show particular promise as a NbS for minimising eutrophication risk. Given the established nutrient pollution issues in Budle Bay, LNNR, this study focuses on constructed wetlands as a NbS to minimise the eutrophication risk from local agricultural runoff. The recent construction of a pilot wetland scheme in Budle Bay 2024 provides an ideal opportunity to evaluate the effectiveness of such interventions.

1.7 Constructed Wetlands

1.7.1 Background

Wetlands act as filters, earning the nickname 'kidneys of the landscape' for their ability to improve water quality by attenuating and processing nutrients (Brix, 1994; Saquib *et al*., 2022). However, agricultural expansion and urban development have caused widespread wetland loss and degradation globally, with approximately 3.4 million km² lost since 1700 (Fluet-Chouinard *et al.*, 2023; Smalling *et al*., 2021).

To address this loss while providing water treatment benefits, constructed wetlands (CWs), also termed reedbeds, have been developed to replicate natural wetland processes for attenuating nutrients from surface runoff (Parde *et al.,* 2021; Turcios *et al*., 2021). However, disagreement exists within the literature regarding CWs' ecological classification. Several studies emphasise their ability to mimic natural wetland processes, while others argue they function more like engineered ponds due to containing ecologically engineered artificial structures (Åhlén *et al.,* 2024; Barman *et al.,* 2024). This classification inconsistency hinders research comparisons when CWs are alternatively termed ponds or excluded from NbS studies.

CWs provide significant co-benefits that enhance their value as NbSs, including biodiversity enhancement through habitat creation for invertebrates, birds, and amphibians, and social benefits such as environment aesthetic improvement, recreational opportunities, and vegetation that can be harvested for craft purposes such as thatching and basket making (Abonyi *et al.* 2024; Cheng *et al*., 2020; Hambäck *et al*., 2023; Masoud *et al.,* 2024). Additionally, CWs mitigate climate change impacts like floods and droughts by storing stormwater and contribute to climate change adaptation through carbon sequestration in vegetation and sediment (Barman *et al*., 2024; Hambäck *et al*., 2023; Hassan *et al*., 2021; Rosentreter *et al*., 2021).

There are two main types of CW: surface flow and subsurface flow *(Figure 1.2).* Surface flow constructed wetland (SFCWs) most closely mimic natural wetlands with water flowing through a series of shallow vegetated ponds in a controlled manner (Goeller *et al.,* 2022; Rodriguez-Dominguez *et al*., 2020). They contain emergent and floating plants alongside diverse microorganisms that facilitate nutrient removal (Deng *et al.,* 2024; Tanner *et al.,* 2022; Wan *et al.,* 2024). They effectively reduce suspended sediment and remove nutrients

from agricultural runoff, while offering simple design with lower construction and operational costs than subsurface wetlands (Hassan *et al.,* 2021; Tanner *et al.,* 2022; Wan *et al.*, 2024).

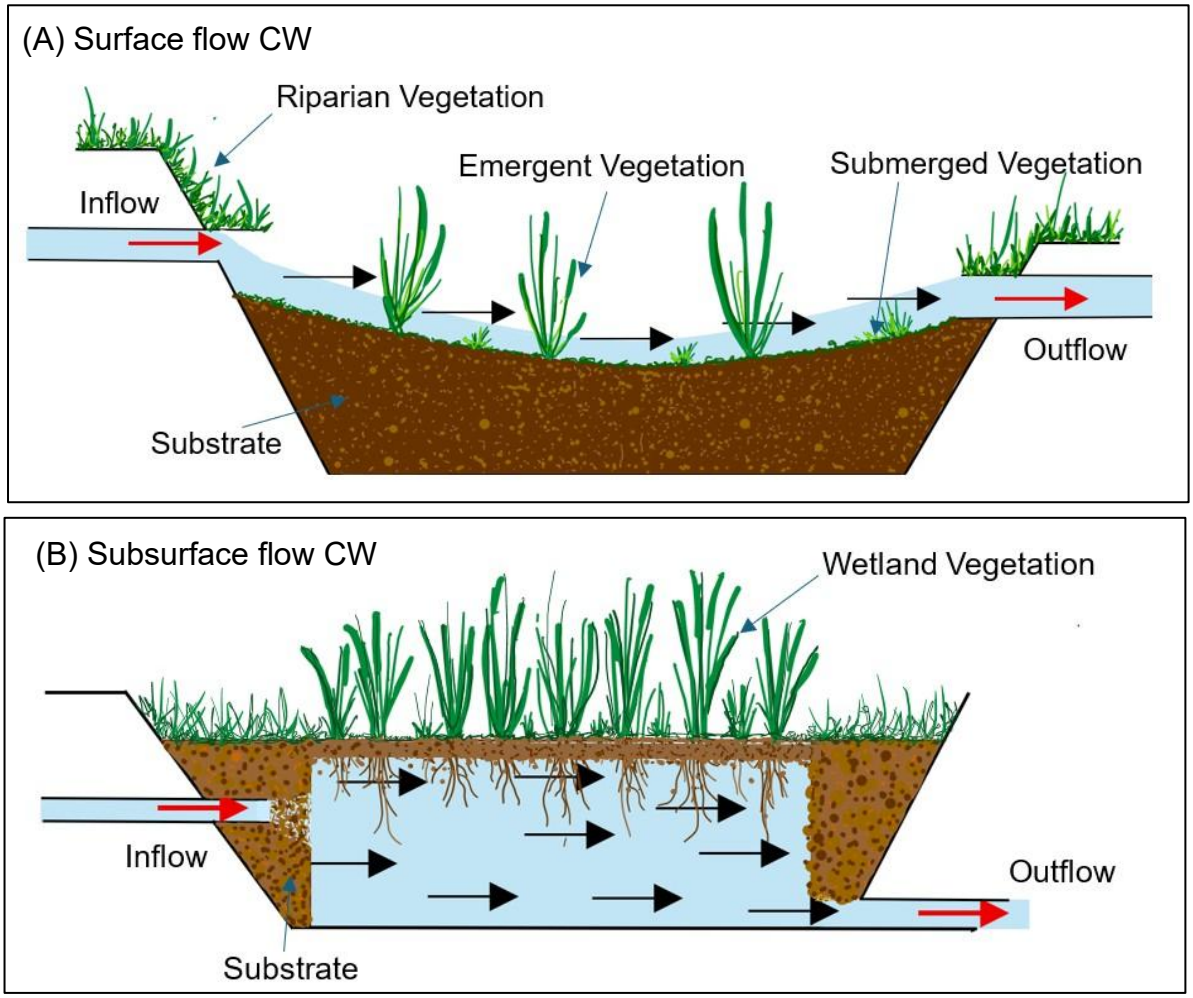


Figure 1.2:(A) Conceptual diagram of a Surface flow constructed wetland (SFCW). Adapted from Tanner et al., *(2022) (B) Conceptual diagram of a Subsurface flow (SSF) constructed wetland. Adapted from Sachdeva* et al. *(2023) and Vymazal (2010).*

Subsurface flow (SSF) wetlands involve the flow of water through a porous substrate such as sand or gravel containing submerged plant roots (Abonyi *et al.,* 2024). Their submerged nature provides enhanced interaction between water and plants, providing a greater surface area for microbial activity and pollutant removal (Alfa *et al.,* 2024). There are three main arrangements of SSF wetlands: vertical, aerated, and floating vegetation. They are each specialised to increase water treatment either by optimising hydraulic distribution, enhancing oxygen transfer to promote aerobic microbial processes, or enhancing the surface area available for plant growth (Abonyi *et al.,* 2024; Landaverde *et al.,* 2024). This study will focus on SFCWs and their ability to treat surface runoff, the term ‘constructed wetlands’ will be used to only refer to this type of wetland.

1.7.2 Pollutant Removal Processes

CWs remove nutrients from agricultural runoff through integrated physical, chemical and biological processes (Dawen and Nabi, 2024; Parde *et al.,* 2021; Tanner *et al.,* 2022). Nitrogenous species (TON and ammonium) are primarily removed from water through sedimentation, adsorption to the soil matrix, microbial processes, and plant uptake (Yousaf *et al.,* 2021).

Planted vegetation creates slow water flow, and extended hydraulic retention time (HRT), promoting gravitational settling of organic matter and adsorption of nitrogenous species to the soil matrix (Hassan *et al*., 2021; Tanner *et al*., 2022; Yousaf *et al.,* 2021). While several studies have found deeper water areas enhance suspended sediment settling, soil matrix adsorption is only a temporary solution of nutrient removal as nitrogen can be remineralised back into solution (Jamieson *et al.,* 2003; Mallin *et al.,* 2012).

Nitrogen has a complex biogeochemical cycle through which microbial processes transform organic and inorganic nitrogen species into less harmful forms, more permanently removing them from runoff (Abonyi *et al.,* 2024; Benham and Mote, 1999; Vymazal, 2007). The three main nitrogen cycle processes responsible for water treatment are nitrification, denitrification, and dissimilatory nitrate reduction to ammonium (DNRA). Nitrification is the biological formation of nitrate or nitrite from ammonium, and it occurs in the aerobic layers of the water column, meaning the process is limited by anoxic conditions (Jamieson *et al.,* 2003).

Denitrification is the main process by which nitrogen is removed and involves the microbial conversion of TON into gaseous nitrogen (Forgrave *et al.,* 2024; Mallin *et al.,* 2012; Tanner *et al.,* 2022, Woodward *et al.,* 2009; Yousaf *et al.,* 2021). This reaction involves enzymatic reduction carried out by anaerobic heterotrophs which substitute oxidised forms of nitrogen (TON) for oxygen as electron acceptors (Vymazal, 2007; Yousaf *et al.,* 2021). Denitrification is most effective under anaerobic conditions, making CWs optimal environments due to their constant nutrient supply that feeds denitrifying bacteria, and their waterlogged conditions that create anoxic environments (Mallin *et al.,* 2012; Tanner *et al.,* 2022).

DNRA, as the name suggests, is the dissimilatory reduction of nitrate (or nitrite) to ammonium (Kim *et al.,* 2011; Vymazal, 2007; Wang *et al.,* 2024). Rather than removing TON from the CW as nitrogen gas, it converts it to ammonium (Deng *et al.,* 2015). The DNRA

process is also strongly controlled by oxygen presence, with the reducing bacteria favouring an environment with relatively high oxygen levels (Quick *et al.,* 2019; Wang *et al.,* 2024).

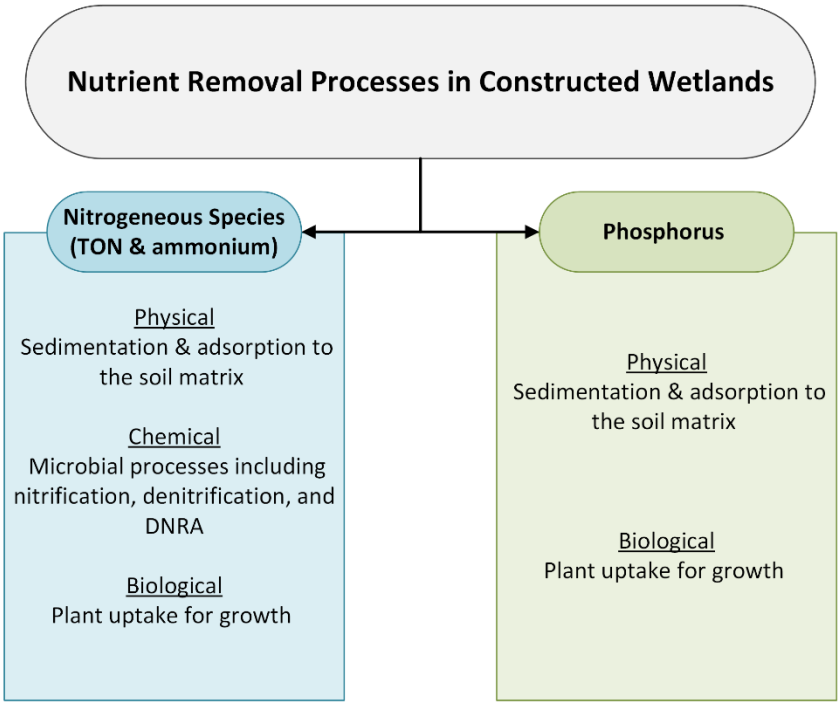


Figure 1.3: Primary physical, chemical, and biological processes involved in the removal of nitrogen and phosphorus from constructed wetlands.

The uptake of nitrogen by plants is also an important removal pathway (Tanner *et al.,* 2022; Yousaf *et al.,* 2021). Plants absorb nitrogen species, converting them into plant biomass that either remineralises or accumulates in wetland soil after decomposition (Abonyi *et al.,* 2024; Tanner *et al.,* 2022). Therefore, this process stores nitrogen within CWs (Jamieson *et al.,* 2003).

The dynamics of phosphorus are less complicated, with the main removal mechanisms being sedimentation and adsorption to the soil matrix, or plant uptake for plant biomass creation (Kim *et al.,* 2011; Mallin *et al.,* 2012). A summary of the nitrogen and phosphorus removal processes in CWs is illustrated in *Figure 1.3.*

By understanding these nutrient dynamics, wetland conditions can be optimised through targeted interventions like oxygen concentration adjustments to maximise removal efficiency, demonstrating why knowledge of underlying biogeochemical processes is essential for effective CW design (Ilyas and Masih, 2018).

1.7.3 Wetland Performance

There is strong agreement within the literature that CWs are effective interventions against agricultural runoff (Hassan *et al.,* 2021; Masoud *et al.,* 2024; Parde *et al.,* 2021). Five studies report that CWs reduce nitrogen, phosphorus, and suspended sediment concentrations by 90-99% (Abonyi *et al,* 2024*,* Barman *et al.,* 2024; Ferreira *et al.,* 2023; Hassan *et al.,* 2021; Parde *et al.,* 2020). However, conflicting results show more modest performance: Wilkins *et al.* (2022) found only 50% nutrient reduction, while Turcios *et al.* (2021) reported poor phosphorus removal despite effective nitrogen treatment. Additionally, Rodriguez-Dominguez *et al.* (2020), found that CW effectiveness declined when treating runoff with higher nutrient concentrations.

1.7.4 Influential Ecological Factors

There are three key ecological factors which influence CW effectiveness, with hydraulic residence time and temperature being more influential than sedimentation (Yousaf *et al.,* 2021).

1. Hydraulic Residence Time (HRT): The retention time in CWs is proportional to the nutrient removal efficiency (Ferreira *et al.,* 2023; Hassan *et al.,*2021). Huang *et al.* (2019) found increased HRT substantially reduced TON and ammonium concentrations. Therefore, nutrient removal can be enhanced by increasing wetland area, creating elongated cells, or implementing multi-cell systems that extend water flow (Tanner *et al.,* 2022; Yousaf *et al.,* 2021). However, multi-cell designs require greater land area and increase costs (Dawen and Nabi, 2024; Hassan *et al.,* 2021). While it is established that additional cells enhance overall treatment performance, current knowledge remains limited regarding how nutrient removal processes are distributed between individual cells within multi-cell systems.
2. Temperature: Nitrification becomes inhibited in water temperatures below 10˚C (Kim *et al.,* 2011; Werker *et al.,* 2002). Microbial activity decreases in low temperatures, while extremely high temperatures can harm microorganisms through oxygen depeletion (Hassan *et al.,* 2021; Nguyen *et al*., 2019). Temperature is also linked to the dissolved oxygen concentration within the CW (Yousaf *et al.,* 2021). CW performance is therefore likely to vary seasonally, with high nutrient removal during warmer months (Matthews *et al.,* 2024; Parde *et al.,* 2020; Wilkins *et al*., 2022). The removal of phosphorus is less affected by temperature as it is absorbed by plants and sediment as opposed to microbial processes (Picard *et al.,* 2005).

3. Sedimentation: CWs gradually accumulate sediment and biomass, causing clogging that decreases treatment efficiency, and causes nutrient remobilisation (Dawen and Nabi, 2024; Wilkins *et al.,* 2022). Therefore, maintenance is required to maintain high nutrient removal efficiency (Deng *et al*., 2024; Parde *et al.,* 2020). Long-term studies on CW operation are currently lacking, meaning sedimentation impacts on CWs remain poorly understood (Gupta *et al.,* 2021).

1.7.5 Wetland Vegetation

Plants are fundamental to CW functioning; promoting sediment settling, nutrient uptake, water oxygenation, and supporting microbial processes (Tanner *et al.,* 2022). CWs contain three main plant types:

1. Emergent plants like reeds (*Phragmites sp.)* and watercress (*Nasturtium officinale*) grow above water in shallow areas (*Figure 1.2)*, with extensive root systems that enhance sediment trapping and nutrient uptake (Abonyi *et al.,* 2024; Márquez-Reyes *et al.,* 2022).
2. Submerged plants, living under the water's surface (*Figure 1.2),* such as pondweeds (*Potamogeton sp.)* oxygenate water but are vulnerable to eutrophication when algae block sunlight (Abonyi *et al.,* 2024; Caputo, 2022).
3. Riparian plants along CWs edges, including great willowherb (*Epilobium hirsutum),* stablise banks, prevent erosion, and rapidly colonise after construction disturbance (Abonyi *et al.,* 2024; Guittonny-Philippe *et al.,* 2015; Nasiri *et al.,* 2024).

Given vegetation's substantial impact on CW effectiveness, understanding post-construction vegetation development is essential for optimising nutrient removal (Abonyi *et al.,* 2024).

1.8 Issues in the Lindisfarne NNR

The Lindisfarne National Nature Reserve (LNNR) is located on the Northumberland coast, approximately 16km south of Berwick-Upon-Tweed, and is a 3500-ha area of tidal flats, dunes, and saltmarshes both within the Holy Island Bay, and Budle Bay to the south (*Figure 1.4)* (Gore *et al.,* 2024; Natural England, 2014). This site is designated under the following conservation statuses:

- Berwickshire and North Northumberland Coast Special Area of Conservation (SAC)
- Northumberland Marine Special Protection Area (SPA)
- Lindisfarne Special Site of Scientific Interest (SSSI)
- Northumberland Shore SSSI
- Lindisfarne SPA and Ramsar

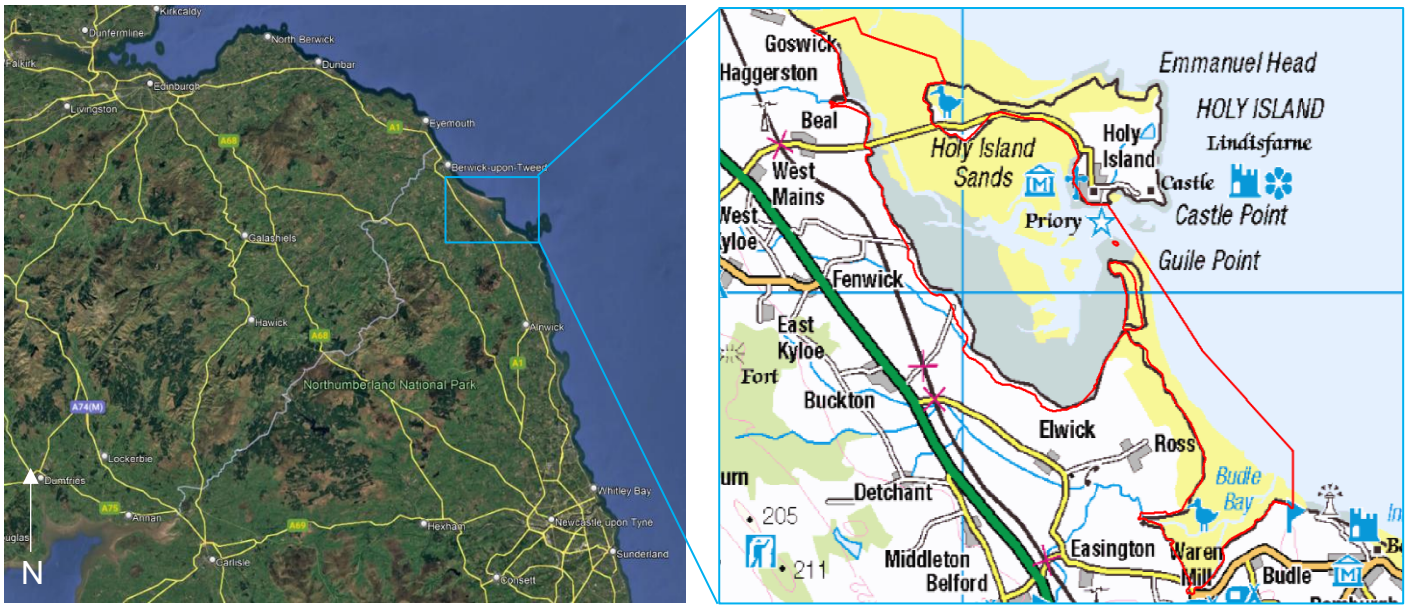


Figure 1.4: Location of the Lindisfarne NNR (red). The Budle Bay section of the LNNR is located to the south-east of Holy Island. Satellite imagery (left) taken from Google Earth Pro (2021), map of the LNNR with red outline taken from Ordnance Survey (2023).

The LNNR has been recognised as an area of potential eutrophication since the early 2000s as part of the OSPAR Comprehensive Procedure (OSPAR, 2003). It was classified as a eutrophic water Nitrate Vulnerable Zone (NVZ) under the EU 1991 Nitrates Directive, and a Sensitive Area (Eutrophic) under the EU 1991 Urban Wastewater Treatment Directive (now the Water Environment Regulations (2017)) (Maier, 2009). The LNNR is also classified as a 'medium priority' Water Priority Area (WPA), a classification for areas where diffuse pollution from agriculture is a significant water quality issue (Natural England, 2016).

Since 2009, extensive blooms of Ulva algae have been recorded on the mud and sand flats of Budle Bay, with between 40ha and 65ha of the bay being covered annually (*Figure 1.5)* (Environment Agency, 2017; Environment Agency, 2021; Mill *et al.,* 2009). These blooms occur seasonally every summer starting in late May and reaching a peak in July when light is not growth limiting (Environment Agency, 2017).



Figure 1.5: Ulva *macroalgae bloom in summer 2018 at the Cockle Ridge side of Budle Bay. Source Crook (pers.comm.).*

This eutrophication, as a result of agricultural runoff, and associated HABs is threatening the biodiversity and species for which the area has been designated to protect. HABs can smother intertidal habitats such as seagrass and saltmarsh, and prevent invertebrates from colonising the anoxic sediment, impacting the critical food supply of over wintering birds. Therefore, Natural England, who manage the LNNR, aim to reduce macroalgae cover across the bay by reducing the nutrient concentrations in the water entering the estuary, and hence improving the water quality (LIFE WADER, 2025b; Wood *et al.,* 2024).

1.9 LIFE WADER Project

The EU-LIFE Water and Disturbance Environmental Restoration (WADER) project is a 5-year, £5.8 million project throughout the Tweed Catchment and Northumberland Coast set to reverse the decline and improve the water quality of more than 49,000 ha of river, coastal and marine habitat. The project, led by Natural England, aims to tackle water quality issues in the area by building several wetland nutrient and sediment trap interventions to help reduce the high levels of diffuse pollution in the streams and drains feeding Budle Bay, and stem nutrient outflow into the estuaries (LIFE WADER, 2025c). As nutrient pollution is a known issue within the Budle Bay catchment, it is an ideal location for pilot testing these NbSs.

Two sites have been chosen by Natural England as priority sites for interventions due to being known inputs of excess nutrients, and their proximity to the LNNR (Renfrey, pers. comm.). These sites will be referred to as Cockle Ridge and Kiln point and are shown in *Figure 1.6.*



Figure 1.6: Location of the Cockle Ridge and Kiln Point intervention sites within Budle Bay (Left) and in relation to the wider Lindisfarne coast (Right). Satellite Imagery taken from Google Earth Pro (2023).

1.10 Cockle Ridge

1.10.1 Drainage ditch (2015)

In 2015, the LNNR team, along with a local cattle farmer who uses the field to the north of the Cockle Ridge drain channel (*Figure 1.7)* created a small drainage ditch to help reduce nutrient runoff into Budle Bay (*Figure 1.8)*. This ditch was left unmaintained and soon became overgrown. No monitoring was undertaken to assess its impact on nutrient dynamics (Renfrey, pers. comm.).



Figure 1.7: Site of Cockle Ridge reed bed and pilot wetlands with surrounding landscape characteristics labelled. Satellite imagery taken from Google Earth Pro (2023).



Figure 1.8: Drainage ditch created in March 2015 by Natural England to mitigate nutrient runoff into Budle Bay. Source Scott (pers. comm.).

1.10.2 LIFE WADER Interventions (2024)

Between the 27th and 30th August 2024, a 3-cell pilot CW intervention was constructed at Cockle Ridge as part of the LIFE WADER project, starting from the location of the existing drainage ditch *(Figure 1.8)* and stretching down towards the upper marsh (*Figure 1.9)*. This project cost approximately £5000 to construct and consisted of site vegetation clearance, regrading and bunding of the area surrounding the uppermost wetland cell (Wetland A) to ensure preferential flow routes, and installation of coir matting logs to act as seepage barriers. Clumps of the *Phragmites australis* reeds from the 2015 drainage ditch were salvaged and placed within Wetland A near the inflow from the upstream drain (*Figure 1.10)* (Renfrey, pers. comm.).

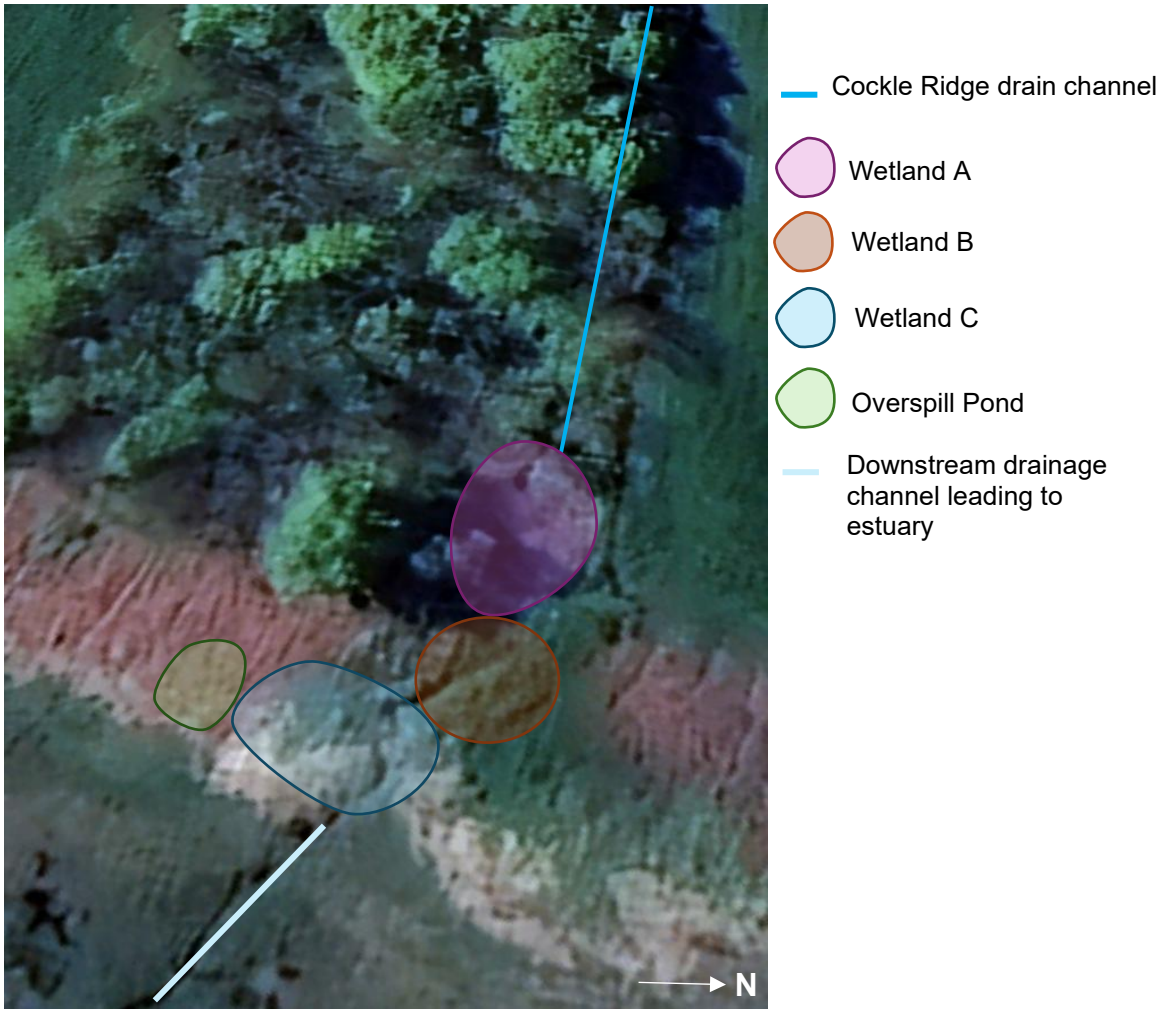


Figure 1.9: Layout of the multi-cell constructed wetland (CW) constructed at Cockle Ridge in August 2024. Satellite imagery taken from Google Earth Pro (2023).



Figure 1.10: Salvaged Pragmites australis *reeds reintroduced into Wetland A near the inflow. Source Renfrey (pers. comm.).*

Between Wetlands A and B, coir matting was installed to the slope to help stabilise the sediment and prevent braiding and preferential flow routes from eroding the soil *(Figure 1.11).* The construction of the two lower cells also had to take the public footpath into account which intersected the proposed location for Wetlands B and C. Therefore, an outflow pipe from Wetland B was run underground, beneath the coastal path, to feed into Wetland C *(Figure 1.12).*



Figure 1.11: Coir matting installed to the slow between Wetlands A and B. Source Renfrey (pers. comm.).



Figure 1.12: Installation of the pipe culvert under the coastal footpath (Left); Boulder used as a baffle on the inflow from the pipe culvert to mitigate direct flow to outflow of Wetland C (Right). Source Renfrey (pers. comm.).

Despite this being a constructed 3-cell system, due to the topography surrounding Wetland C, an overspill pond naturally developed at the base of the intervention, making it into a 4-cell system (*Figure 1.13)*. Therefore, the Cockle Ridge pilot intervention consists of 3 purpose-built cells (Wetlands A-C) and one overspill pond. However, the existence of the overspill pond is reliant on there being sufficient water within the system due to its shallow nature.



Figure 1.13: The three constructed wetland cells (A, B,& C) at Cockle Ridge (Left); The shallow overspill pond which has formed below Wetland C (Right). Source Renfrey (pers. comm.).

The success of this pilot multi-cell CW in mitigating nutrient pollution to Budle Bay could drive wider implementation of similar NbSs across the LNNR to address diffuse pollution challenges.

1.11 Proposed Constructed Wetlands at Kiln Point

1.11.1 Kiln Point Nutrients

While there is some agricultural activity near the Kiln Point field drain, the main source of nutrients is believed to originate from an underground aquifer and natural spring line in the area. This remains speculation and is yet to be confirmed on-site. Natural England also speculate that nutrient enrichment in the field drain could results from off-site infiltration, surface run off and groundwater flows. Additional potential sources include amenity grass from local golf courses and caravan parks, an old drainage culvert from Budle Hall, and minor local agricultural drainage (Renfrey, pers comms.).

1.11.2 Outflow Diversion (2016)

In 2016, the outflow of the Kiln Point field drain underwent modifications to divert the surface runoff away from the bridle way through an existing underground culvert *(Figure 1.14).* This diversion was most obviously observed through the change in location of macroalgae growing at the outflow, after diversion macroalgae is no longer found at the base of the bridle path *(Figure 1.14)* (Craggs, pers. comm.; Crook, pers. comm.).



Figure 1.14: Outflow of the Kiln Point field drain (blue) in 2006 before diversion (Left) and in 2018 after diversion (Right). Where the field drain and surface runoff enters the intertidal area is illustrated in orange, with macroalgae growth clearly visible in these locations. The underground culvert installed in 2016 is illustrated in yellow. The bridle way is shown in red. Sourced from Google Earth Pro (2006) and Google Earth Pro (2018).

1.11.3 Proposed Intervention (2025)

A CW intervention is proposed for constructed at Kiln Point in late 2025, providing the pilot project at Cockle Ridge proves successful. The proposed intervention is illustrated in *Figure*

1.15 and is expected to consist of a 5-cell system, associated land works and vegetation clearance; anticipated to cost approximately £40,000. Construction may deviate from the drawings depending on ground conditions. WADER has not set specific improvement targets but seeks environmental betterment evidenced through reduced opportunistic macroalgae at the Kiln Point outflow and lower nutrient concentrations during monitoring (Renfrey, pers comms.).

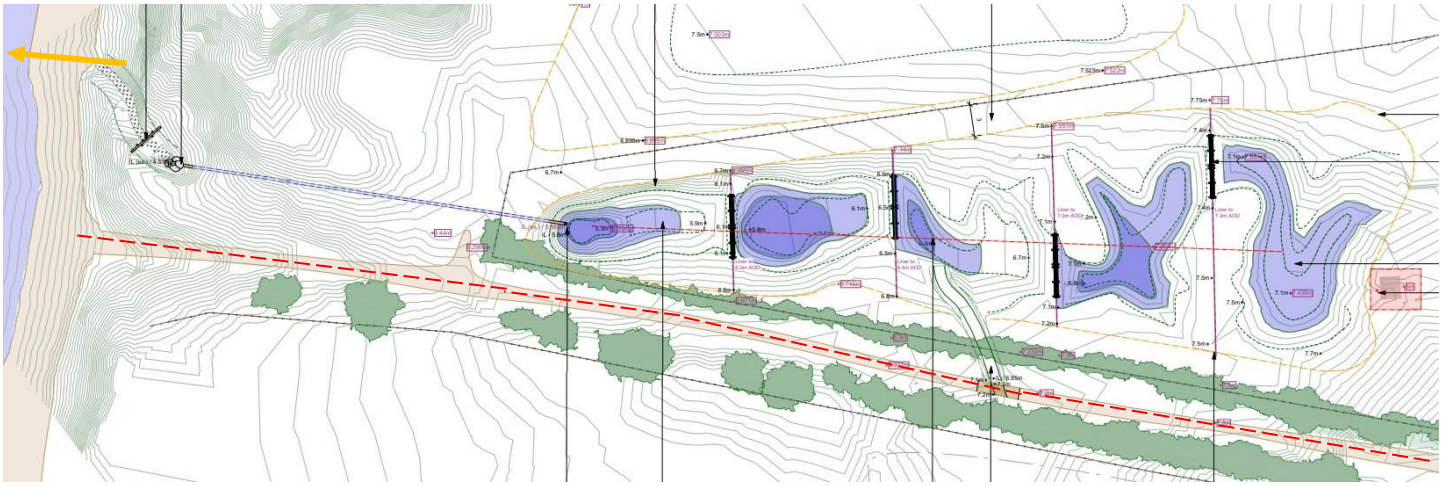


Figure 1.15: Preliminary technical drawings of the proposed Kiln Point wetland intervention created in January 2024 by Blue Green Studio. The bridle way is illustrated by the red dashed line and the outflow into the estuary in orange. Provided by Welch (pers. comm.)

1.12 Research Aims and Questions

Therefore, this study aims to evaluate the effectiveness of constructed wetlands (CWs) as interventions against estuarine nutrient enrichment, advancing understanding of multi-cell treatment processes and post-construction wetland evolution.

To achieve this, a case study system is required where past nutrient concentrations are documented, post-construction changes can be assessed, and future treatment efficiency can be forecasted. Therefore, this study focuses on two key nutrient sources at Budle Bay: Cockle Ridge and Kiln Point. By analysing past and present nutrient concentrations, assessing the nine-month treatment performance of the 2024 Cockle Ridge CW intervention, and forecasting the potential effectiveness of the proposed Kiln Point CW, this research will provide evidence-based recommendations for future CW implementation and monitoring strategies within the Lindisfarne National Nature Reserve.

Limited historical data is available at both field sites, constraining the investigation of past nutrient dynamics. For this study, ‘historical’ data encompasses any monitoring data recorded before this research commenced that evidence nutrient changes pre- and post-intervention. Given the substantial novel data generated on the Budle Bay wetlands, the 'present' analysis has been subdivided into detailed subsections for comprehensive system performance.

To address this aim I have 3 research questions:

1. What are the historic baseline nutrient concentrations at Cockle Ridge and Kiln Point, and how have conditions changed since the Cockle Ridge constructed wetland was installed in August 2024? (Past)
2. How effectively is the Cockle Ridge pilot constructed wetland reducing nutrient loads to Budle Bay nine months post- construction? (Present)
    - Is treatment performance consistent across the 4-cell wetland system?
    - What do the nutrient dynamics indicate about the wetland processes driving the observed treatment efficiency?
3. Based on comparative nutrient loading and observed treatment performance at Cockle Ridge, and current site conditions at Kiln Point, what is the predicted effectiveness of the proposed Kiln Point constructed wetland, and what are the recommendations for its implementation and monitoring strategy? (Future)

2 Methods

2.1 Field Sites

To assess the effectiveness of the Cockle Ridge pilot intervention at reducing nutrient runoff and monitor the nutrient concentrations around the proposed CW site at Kiln Point, water and sediment sampling of the two field sites in Budle Bay, LNNR was conducted between the 14th -16th May 2025 (*Figure 2.1)*. Permission was obtained from Natural England (NE) and local landowners to undertake fieldwork at these sites. All work adhered to NE's risk assessment guidance using a buddy system, with regular location updates to Natural England and University of Edinburgh staff. Sampling at both sites was timed according to tides to ensure safe access to low-elevation areas at Cockle Ridge and prevent tidal interference with drainage channel sediment sampling.

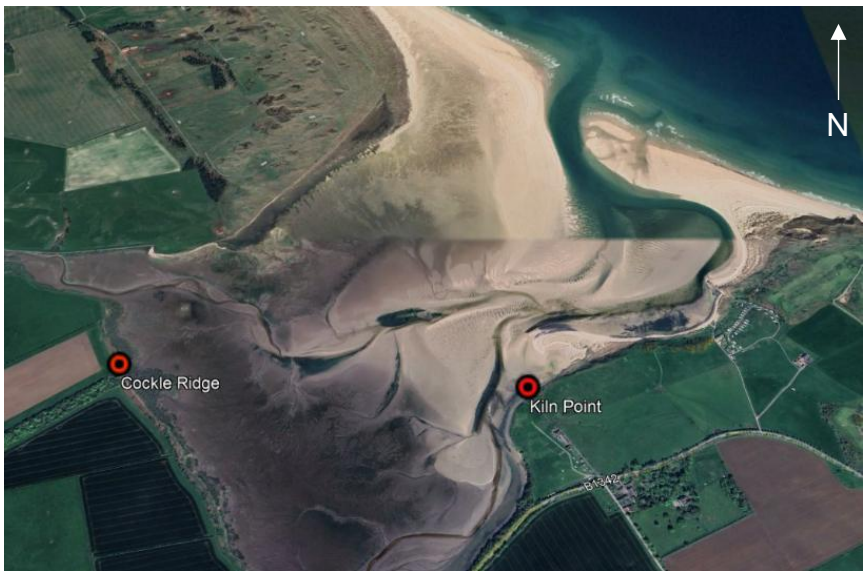


Figure 2.1: Location of the Cockle Ridge and Kiln Point field sites within Budle Bay, Lindisfarne National Nature Reserve (LNNR). Satellite Imagery taken from Google Earth Pro (2023).

2.2 Fieldwork Experimental Design

2.2.1 Cockle Ridge

This was the primary sampling site with both water, and sediment samples taken from 6 locations. At Cockle Ridge, 34 water samples were collected, 5 replicates from the up- and downstream areas and 6 replicates for each of the 4 cells, shown in *Figure 2.2*. Water samples were taken using a 20ml collection tube taped to a bamboo pole which allowed for water to be collected while maintaining a safe distance from the wetland. For each sample, 40ml was taken from the wetland and filtered, using a 0.45 µm filter attached to a 25ml syringe. This took 2 repeats to filter the whole sample through the 25ml syringe, whilst reusing the filter. Filtering the samples was necessary to remove suspended matter, algae and other microorganisms that may cause changes to the concentrations of dissolved nutrients before the sample is analysed; a new filter was used for each sample (European Commission, 2009). However, for some sites where suspended sediment was higher, a second filter was used. The filtered samples were stored in 50ml skirted centrifuge tubes which were labelled with a unique identification code.



Figure 2.2: Water and sediment sample locations at the Cockle Ridge pilot constructed wetland intervention. Satellite imagery taken from Google Earth Pro (2023).

At Cockle Ridge, 16 sediment samples were taken by digging a small quantity of sediment from the base of the wetland cell or drainage channel using a trowel taped to an extendable metal pole. This allowed sediment to be collected while minimising the disturbance to the wetland. Three sediment samples were taken within each wetland cell and from within the downstream drainage channel and two water samples were taken from both the left and the right sides of the intervention, to compare nutrient levels of the wetland against surrounding soil (*Figure 2.2)*. No sediment samples were collected from the upstream drain channel as this was entirely covered by a tree canopy, contributing substantial organic matter (e.g., leaves) to the channel, skewing nutrient results. The collected sediment was stored in 50ml centrifuge tubes for transportation, each labelled with a unique identification code.

2.2.2 Kiln Point

The above methods were repeated for water and sediment sampling at Kiln Point. Seven water samples were collected, three taken from the field drain and four taken from the downstream outflow (*Figure 2.3)*. Additionally, three sediment samples were taken from the field drain; no sediment was taken from the downstream outflow as it flowed over beach shale and rock. All samples were labelled with a unique identification code.



Figure 2.3: Water and sediment sample locations at the location proposed for the Kiln Point constructed wetland. Satellite imagery taken from Google Earth Pro (2023).

The sediment and filtered water samples from both sites were frozen at -18 ˚C to preserve the dissolved concentrations of nutrients within them. To transport the samples back to the University of Edinburgh for laboratory analysis the samples were packed into coolers and insulated bags packed with ice to keep them frozen. The samples were then stored in -18˚C freezers until analysis.

2.2.3 Observations

At Cockle Ridge, photos were taken around the sampling locations to record plant species present. Plant species were then identified using Field Studies Council identification guides covering waterside flowers, common grasses, grassland plants, floodplain meadows, and common water plants (Gardener and Roberts, 2010; Gowing *et al.,* 2010; Gulliver *et al.,* 2002; Orton *et al.,* 2000; Simon and Harper, 2024).

2.3 Sample Processing

Each sample was processed according to the below protocols; for water samples 25% of samples were run in duplicate, and for sediment samples 100% of samples were extracted a second time due to the higher sources of error reflected within these samples *(Section 2.4).* Sample duplicate concentrations were then averaged at the end of calculations *(Section 2.5.2).*

2.3.1 Water Samples

2.3.1.1 Total Oxidised Nitrogen

To measure the sum of the total oxidised nitrogen (TON), a mixed reagent made of a combination of Greiss reagent (comprised of sulphanilamide solution and N-(1-Naphthyl)ethylenediamine dihydrochloride) and Vanadium (III) chloride (VCl_3) reagent is used as described in Grasshoff *et al.* (1983) and García-Robledo *et al.,* (2014). This method has high accuracy and specificity and a low detection limit without using complex procedures. TON should react with the mixed reagent to form a pink coloured complex, known as the Greiss Reaction, with the intensity of the pink colouration being directly proportional to the TON concentration present.

A series of 9 working nitrate standards were used, consisting of one blank and 8 dilutions of 200 µM KNO_3 ranging between 0.2 µM NO_3 and 30 NO_3. All standards and blanks were

prepared using deionised water produced by the Siemens Ultra Clear RO unit. Completing a parallel analysis of NO_3 standards on the spectrophotometer will help to determine the TON concentration within the samples and identify any contamination (García-Robledo *et al.,* 2014).

The water samples were defrosted and 10ml was measured out into a 15ml centrifuge tube. Sample concentrations that exceed 30 µM are unable to be accurately read in the spectrophotometer as this is the upper limit for the measurement of TON**.** Therefore, samples with especially high concentrations were diluted to obtain an accurate reading (Grasshoff *et al.,* 1983; García-Robledo *et al.,* 2014). For water samples it was found that a dilution rate of x10 (1ml of sample plus 9ml of deionised water) or x30 was required dependent on the site from which the sample was taken. The water samples taken at Kiln Point and those taken from the upstream drain channel and wetland A at Cockle Ridge required a x30 dilution, while all other samples were diluted x10 to be within the range of the prepared standards.

To each of the 10ml diluted samples, plus the 9 standards, 1ml of the mixed reagent was pipetted, the samples were shaken and placed into a 60˚C water bath for 45 minutes *(Figure 2.4).* During this time, the samples should have undergone the colour change *(Figure 2.4).* Samples were allowed to cool to room temperature before measuring in the Shimadzu UV-Vis UV-1900i spectrophotometer using a 1cm path length quartz cuvette and the

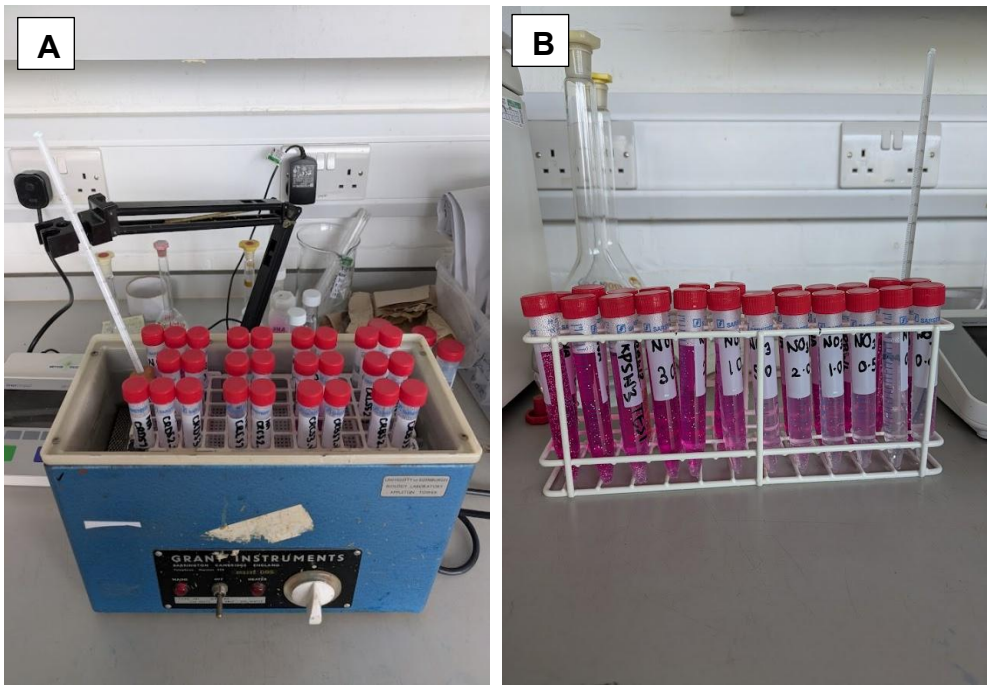


Figure 2.4: (A) Samples and standards in the water bath at 60˚C for 45 minutes (B) Samples and standards following the water bath treatment being left to cool, the amount of colour change undergone by the sample is related to the concentration of total oxidised nitrogen present.

concentration (µmol L^{-1}) measured at 540nm, with the cuvette rinsed with deionised water between samples.

2.3.1.2 Phosphorus

To determine the concentration of soluble phosphorus, including labile phosphorus species such as orthophosphate, within the water samples the molybdenum blue method, as detailed by Nagul *et al.* (2015), was followed. The University of Edinburgh technical services team prepared molybdenum mixed reagent, consisting of ammonium molybdate solution, antimony potassium tartrate solution, sulphuric acid solution, and ascorbic acid solution. This reagent will turn blue when added to a sample containing phosphorus with the depth of colour dependent on the concentration present (*Figure 2.5)*.

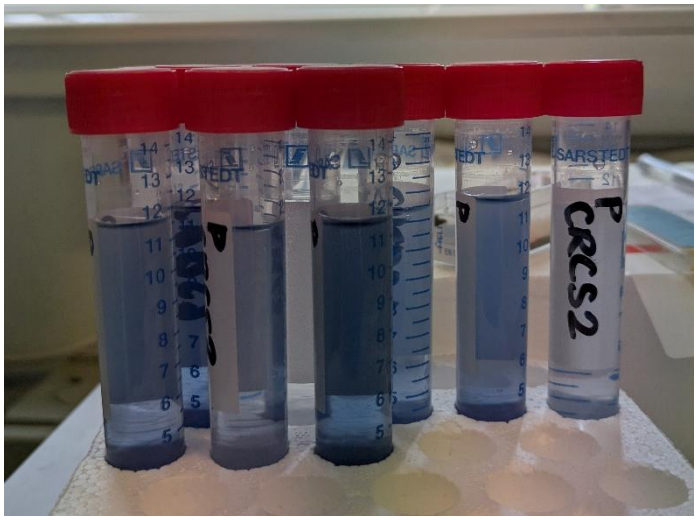


Figure 2.5: Colour change of samples following the addition of molybdenum mixed reagent to determine phosphorus concentration. The amount of colour change undergone by the sample is related to the concentration of phosphorus present.

A series of 6 working standards comprised of one blank and 5 dilutions of 10 mg L^{-1} P stock solution, provided by the University of Edinburgh lab technicians, were created with concentrations ranging between 0.02 mg L^{-1} and 0.1 mg L^{-1}. All samples were found to be within the concentration range of the standards and therefore no dilutions were necessary. To the water samples, plus 6 standards, 1ml of the molybdenum mixed reagent was added and swirled to mix. After 5 minutes the concentrations (mg L^{-1}) were measured at 880nm using the spectrophotometer.

2.3.1.3 Ammonium

Testing the water samples for Ammoniacal Nitrogen, filtered as N (mg L^{-1}) (hereafter referred to as ammonium) was carried out with help from the technical services team who prepared

all reagents and standards and set up the equipment. The analysis was carried out using a SEAL Analytical XY-2 Sampler connected to a Flow injection analysis system, feeding into a spectrophotometer. All water samples, including blanks and replicates, were placed in designated labelled positions on sample trays, the auto analyser uses a mobile probe to aspirate a small portion of each sample and inject it into the carrier stream. Each sample is mixed with the reagents inside the coils of the Flow injection analysis system. The reagent turns green in samples containing ammonium, with the depth of colour being equivalent to the concentration of ammonium present. The concentration of ammonium (mg L^{-1}) was measured in the spectrophotometer at 660nm. The auto-analyser ran until all samples had been processed.

2.3.1.4 pH

The pH for water samples was read using a SevenCompact S220 pH benchtop meter. The pH probe was calibrated and rinsed with deionised water and gently dried with paper tissue using a blotting motion. The pH probe was immersed in a sample, pressing read to start a measurement and waiting for a stable value before recording. The probe was washed with deionised water between each sample.

2.3.2 Sediment Samples

2.3.2.1 Total Oxidised Nitrogen

To extract TON from the sediment samples, 3g was measured into a measuring boat on a Sartorius B120S balance to 4 decimal places. This was added to a 50ml centrifuge tube with 30ml of 2M $KCl_{(aq)}$ provided by the University of Edinburgh lab technicians. This was repeated for all 19 sediment samples which, along with a blank comprising of only $KCl_{(aq)}$, were shaken for 30 minutes using an end-over-end shaker at 30 RPM. All 20 samples were then filtered through Whatman no. 42 filter paper and 10ml was collected into 15ml centrifuge tubes which were labelled and stored in the fridge before analysis. The same set of 9 standards from the water nitrate analysis were used to run the sediment samples against.

TON was determined within the sediment extractions using the same mixed reagent and method for the water TON analysis to produce a Greiss Reaction. For the sediment samples it was found that a dilution rate of x5 was necessary for all samples to be within the range of the prepared standards. Samples were measured in the spectrophotometer at 540nm for TON concentration (µmol L^{-1}).

2.3.2.2 Phosphorus

The same molybdenum blue method, from Nagul *et al.* (2015), used in the water phosphorus analysis was followed. The set of 6 standards for the water phosphorus analysis were used to test the sediment samples against.

Using the balance, 0.5g of sediment was measured into a measuring boat to 4 decimal places. This was added to a 50ml centrifuge tube with 20ml of 0.11 M acetic acid solution provided by the University of Edinburgh lab technicians. This was repeated for all 19 sediment samples which, along with a blank comprised of only the acetic acid solution, were shaken for 20 minutes on an end-over-end shaker at 30 RPM. All 20 samples were filtered through Whatman no. 42 filter paper and 10ml was collected into 15ml centrifuge tubes which were labelled and stored in the fridge before analysis.

All sediment extracts were diluted by a factor of 5, which allowed all sample concentrations to fall within the standards' range. Using a pipette, 1ml of the molybdenum mixed reagent was added and samples swirled to mix. After 5 minutes the concentrations (mg L^{-1}) were read using the spectrophotometer at 880nm.

2.3.2.3 Ammonium

Available ammonium was extracted from the sediment by following the method used for TON extraction in *section 2.3.2.1*. Analysis of the sediment extracts for ammonium (mg L^{-1}) was carried out using the same procedure as the water samples and the concentration of ammonium (mg L^{-1}) was measured in the spectrophotometer at 660nm

2.3.2.4 pH

To test sediment samples for pH, 5g of sediment was measured into a measuring boat on the balance to 4 decimal places and mixed with 20ml of 0.01M $CaCl_{2\,(aq)}$ in a 50ml centrifuge tube. Samples were shaken for 15 minutes using an end-over-end shaker at 30 RPM. The pH of the extraction was then measured following the method in *section 2.3.1.4.*

2.3.2.5 Dry Mass

Using the balance, 2g of each sediment sample was measured into a measuring boat to 4 decimal places to determine the wet mass. The sediment was placed in labelled aluminium foil dishes and placed into a 70˚C oven for 24 hours, the oven-dried sediment was reweighed

to establish its dry mass. This value was used in the concentration calculation in *section 2.5.1.*

2.4 Error

Throughout the analysis process there is potential for error within the results due to the use of several pieces of equipment and measuring volumes by hand. It is important to recognise potential error and account for it, but it does not negate the general trends observed within the data.

Potential sources of Error:

- Sartorius B120S balance
- Shimadzu UV-Vis UV-1900i spectrophotometer
- SEAL Analytical XY-2 Sampler
- Flow Injection Analyser
- Chemical Error: degradation of mixed reagent and contamination of extraction solutions.
- Human Error: including pipetting and variations in waiting times between analysis

In order to mitigate error in concentration analysis, blanks were run in tandem with the water and sediment samples. All blanks were within the range of 0-0.007 mg L^{-1} except for those used in the sediment TON and ammonium analysis, with these blanks ranging between 0.4-1.5 mg L^{-1}. As the spectrophotometer readings for the nutrient concentrations of the blanks should have been 0 mg L^{-1}, any readings above zero highlights potential contamination within the process. The readings were above zero, suggesting that there was contamination during the extraction process with the 2M $KCL_{(aq)}$ solution. Therefore, the final nutrient concentrations were corrected to account for this error as highlighted in *section 2.5.2.*

2.5 Concentration Calculations

2.5.1 Concentration Conversions

The spectrophotometer concentration readings for water and sediment TON were converted from µmol L^{-1} to mg L^{-1} using *Equation 1.*

Equation 1: Conversion needed to change total oxidised nitrogen concentrations from µmol L^{-1} to mg L^{-1} to allow for comparisons against the other two nutrients.

$$\text{mg L}^{-1} = \mu mol\ \text{L}^{-1} \times (Molecular\ Weight \div 1000)$$

Molecular Weight:
Nitrate: 14g/mol

Following conversion, the final TON, phosphorus, and ammonium concentrations for both water (mg L^{-1}) and sediment (mg Kg^{-1}) samples were calculated using *Equation 2* and *Equation 3* respectively.

Equation 2: Equation needed to convert the water concentrations of TON, phosphorus, and ammonium taken from the spectrophotometer or the conversions using Equation 1, into their final concentrations by accounting for the dilution factors used.

$$Final\ Water\ Concentration\left(\text{mg L}^{-1}\right) = UV - Vis\ Concentration\left(\text{mg L}^{-1}\right) \times Dilution\ Factor$$

Dilution Factors
TON: x10 or x30 dependent on sample location (see section 2.3.1.1)
Phosphorus: no dilution necessary
Ammonium: no dilution necessary

Equation 3: Equation needed to convert the sediment concentrations of TON, phosphorus, and ammonium taken from the spectrophotometer or the conversions using Equation 1, into their final concentrations by accounting for the dilution factors, extraction volumes, and extraction mass used.

$$Final\ Sediment\ Concentration\ (\text{mg}\ kg^{-1}) = \frac{UV\text{–}Vis\ Concentration\left(\text{mg L}^{-1}\right) \times Dilution\ Factor \times Extraction\ Volume\ (ml)}{\frac{Dry\ Mass\ (g)}{Wet\ Mass\ (g)} \times Extraction\ Mass}$$

Dilution Factors & Extraction Volume & Extraction Mass
TON: x5; 30ml of 2M KCl solution, 3g
Phosphorus: x5; 20ml of 0.11M Acetic Acid solution, 0.5g
Ammonium: no dilution necessary; 30ml of 2M KCl solution, 3g

2.5.2 Blank Corrections

Blanks were run with each batch of samples and used for comparison against samples. To avoid an over estimation of nutrient concentrations, the samples were corrected using the average blank concentration for each nutrient-type analysis, shown in *Table 1*. This was calculated by taking the blank correction value from the final concentrations calculated using *Equations 2 & 3.*

Table 1: Blank correction values used to correct the sample final concentrations to account for the potential contamination during sample processing.

Blank Sample Type	Total Oxidised Nitrogen (TON) blank correction value	Phosphorus blank correction value	Ammonium blank correction value
Water (mg L^{-1})	0	0.0026	0
Sediment (mg kg^{-1})	0.9955	0.0065	0.9227

After all the samples had been corrected against the blanks, the sample replicates were than averaged to create mean TON, phosphorus, and ammonium concentration for each water and sediment sample. These mean nutrient concentrations for each sample were then used for the statistical analysis (*section 2.7*).

A flow schematic summarising the sample processing for both water and sediment samples is provided in *Figure 2.6.*

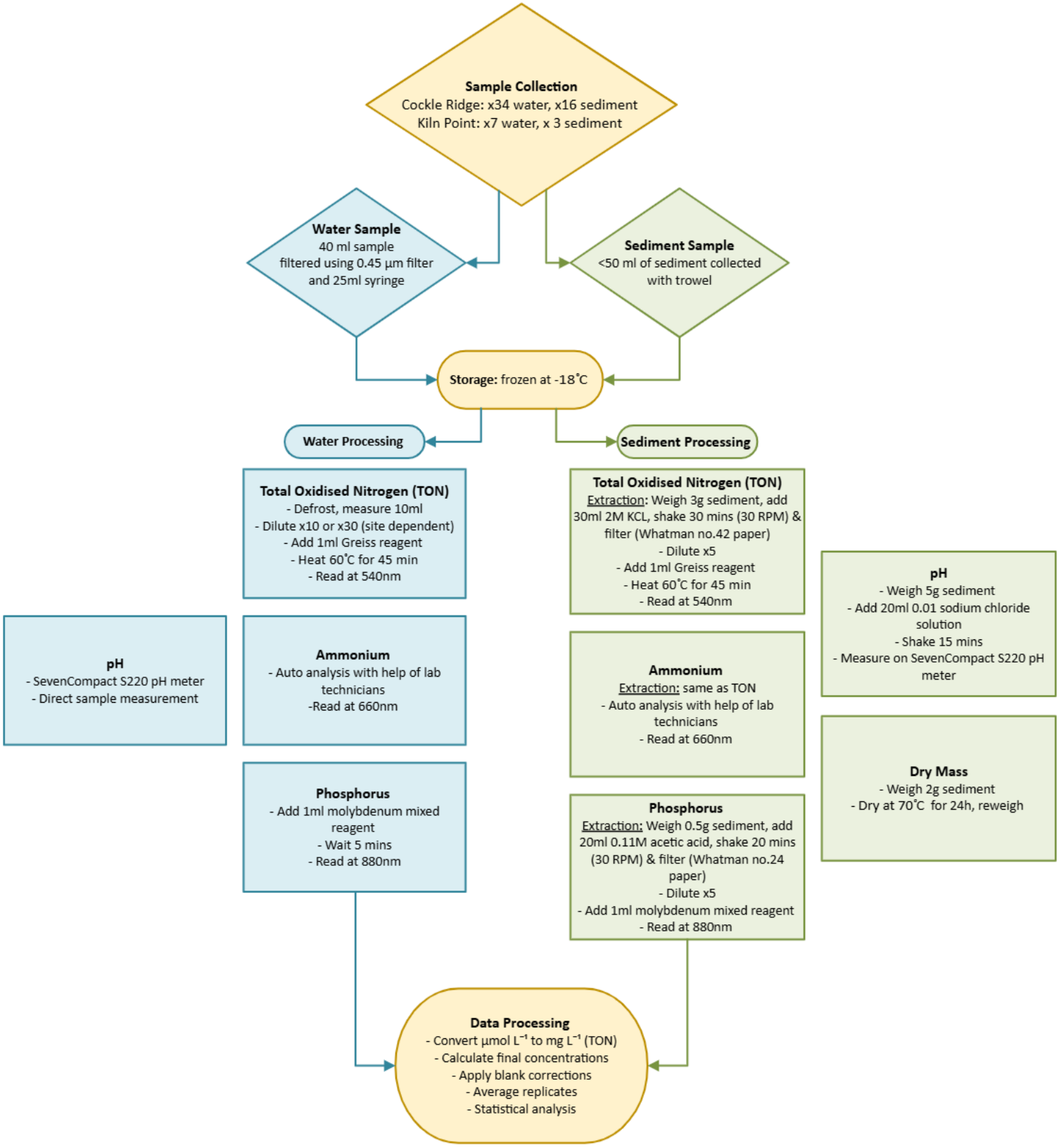


Figure 2.6: Flow schematic of processes involved in sample analysis. Each water sample was analysed for 3 nutrients (TON, phosphorus, and ammonium) and pH, and each sediment sample was analysed for 3 nutrients (TON, phosphorus, and ammonium), pH, and the dry mass of sediment necessary of the concentration calculations.

2.6 Secondary Datasets

Two external datasets were used for comparison against the findings from this study, to understand how the nutrient concentrations at Cockle Ridge and Kiln Point have changed over time.

1. Environment Agency Data (2016-2017)

   Between 2016-2017 the Environment Agency conducted water sampling twice a month at 12 coastal sites in the LNNR, including Cockle Ridge and Kiln Point, as part of a wider sampling effort to determine areas of high nutrient concentration. The method used was the standard WFD surveillance water sampling method and the samples were analysed by the National Lab Service (European Commission, 2003). Samples were analysed for total oxidised nitrogen (TON), Ammoniacal Nitrogen, and Orthophosphate (mg L^{-1}), comparable to the testing from this study. Data was provided by Crook (pers. comm.).
   The data from the following two Environment Agency monitoring locations was used to assess historic nutrient concentrations:

   - Cockle Ridge Reed Bed Outflow: Monitoring location 42100238
   - Drain At Kiln Point: Monitoring location 42100204

2. Natural England Monitoring Dataset (2024-2025)

   Since August 2024, Natural England has monitored nitrate, ammonium, and phosphorus concentrations using a Hanna photometer. Monthly monitoring occurred at Cockle Ridge, while Kiln Point was sampled only three times during this period; this inconsistent sampling frequency between sites may create challenges when comparing data. While nitrate concentrations do not directly represent TON concentrations because they exclude nitrite, nitrite typically constitutes a relatively small fraction of oxidised nitrogen in CWs (Kadlec, 2005). Consequently, nitrate concentrations measured by Natural England serve as a reasonable approximation of TON concentrations at these sites. Data was provided by Renfrey (pers. comm.).

2.7 Statistical Analysis

Statistical analysis was performed in R studio (R version 4.5.1; R Core Team, 2025) using the programming language R. To prepare data for processing, negative values and values 1.5 times higher or lower than the IQR from the upper quartile and lower quartile respectively were filtered out from the mean nutrient concentration data. This method of removing outliers was used as the data was not normally distributed (Tukey, 1977). This filtered out most phosphorus water concentrations which were negative following the blank correction.

Data analysis used the tidyverse collection of packages (Wickham *et al*., 2019). Analysis included displaying average nutrient concentrations at each sample location, grouped by nutrient type, and comparisons made against historic data. Error bars were included to illustrate the variation within the data. The correlation between nutrient type and wetland number was determined using a scatter plot fitted with a Loess trend line to reveal the trends underlying the scattered sample location replicates (Cleveland, 1979). A Spearman's rank correlation test, more suitable for non-normally distributed data, was undertaken with the level for statistical significance set at $P<0.05$ (Zar, 2005).

For each successive wetland cell (A, B, C, and Overspill), the mean concentration of TON at each wetland, and the concentration entering from the upstream drain was used to calculate the % reduction in TON per wetland cell, following the same method as Robotham *et al.* (2021). To predict the reduction efficiency of a fifth wetland cell, a simple trend analysis was used, recognising the diminishing reduction in TON with increased wetland number. This follows Land *et al.* (2016) who demonstrated in a systematic review of 203 wetlands that removal efficiency decreases with increasing hydraulic loading rates and becomes very low at low nutrient concentrations, known as the principle of diminishing marginal returns. To extrapolate wetland performance to a five-cell system, a conservative assumption was made based on this principle (Land *et al.,* 2016): the % reduction in TON would increase by only 30% of the reduction achieved by the fourth cell, reflecting that the remaining nutrient fraction becomes increasingly difficult to remove.

3 Results

3.1 Assessing Historic Nutrient Concentrations in Budle Bay

Between November 2016 and November 2017, monitoring by the Environment Agency found that total oxidised nitrogen (TON) was the nutrient in highest concentration recorded in runoff at both Cockle Ridge and Kiln Point outflows *(Figure 3.1).* At Cockle Ridge, TON had a mean concentration of 4.47 mg L^{-1} (*Standard Deviation (SD)* = 1.88) whereas phosphorus and ammonium had mean concentrations of 0.079 mg L^{-1} (*SD* = 0.041) and 0.032 mg L^{-1} (*SD* = 0.009) respectively. At Kiln Point, TON had a mean concentration of 4.68 mg L^{-1} (*SD* = 0.288) whereas phosphorus and ammonium had mean concentrations of 0.016 mg L^{-1} (*SD* = 0.029) and 0.03 mg L^{-1} (*SD* = 0.002) respectively.

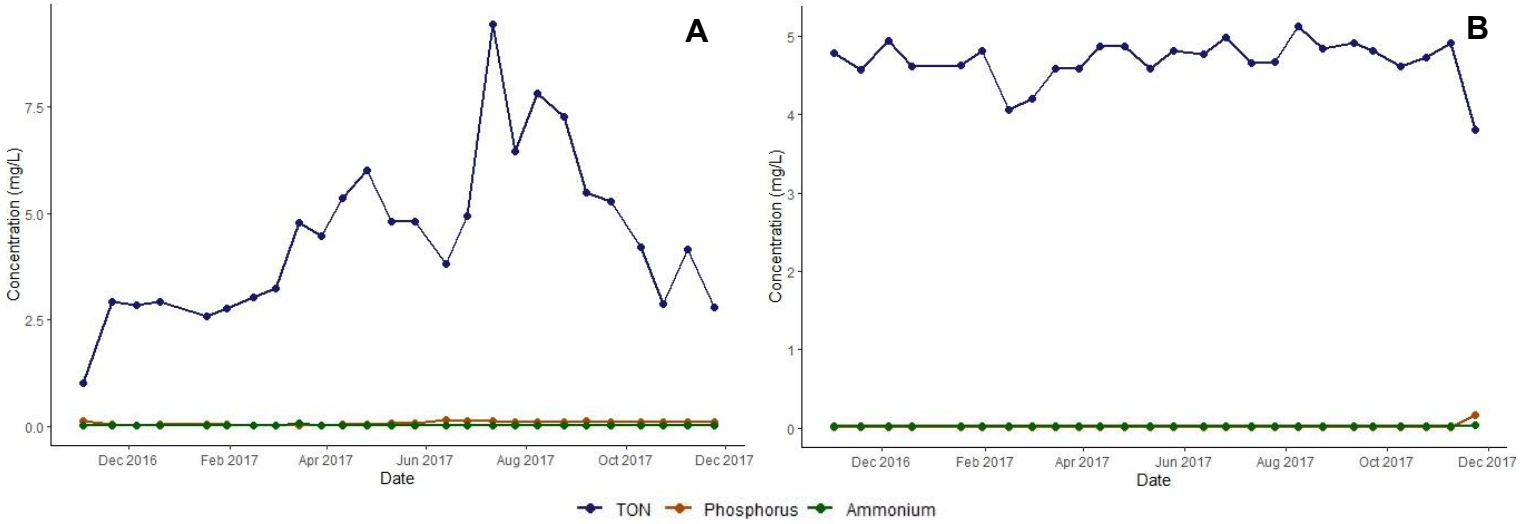


Figure 3.1: Environment Agency data showing changes in TON, phosphorus and ammonium between November 2016 and November 2017 at (A) Cockle Ridge reedbed outflow (EA monitoring point 42100238) and (B) Drain at Kiln Point (EA monitoring point 42100204). Data provided by Crook (pers. comm.).

Natural England monitored Cockle Ridge and Kiln Point sporadically between August 2024 and May 2025. Nitrate, a component of TON, was found to be the nutrient in highest concentration at both Cockle Ridge and Kiln Point. At Cockle Ridge, the mean nitrate concentration across all sample locations was 3.52 mg L^{-1} (*SD* = 1.41) with mean phosphorus and ammonium concentrations of 0.36 mg L^{-1} (*SD* = 0.53) and 0.37 mg L^{-1} (*SD* = 0.59) respectively *(Figure 3.2).* The upstream sampling location has the greatest mean concentration of nitrates at 4.8 mg L^{-1} *(SD* = 0.58), and the Overspill Pond had the lowest mean of 2.6 mg L^{-1} (SD = 1.5). Conversely, for the mean concentration of phosphorus and ammonium, Wetland C had the greatest concentration with 0.733 mg L^{-1} (*SD* =0.351) and

0.7 mg L^{-1} (*SD* =0.906) respectively with the lowest mean concentration of phosphorus observed Upstream (0.15 mg L^{-1} (*SD* = 0.19) and the lowest mean concentration of ammonium observed in Wetland B (0.15 mg L^{-1} (*SD* = 0.3).

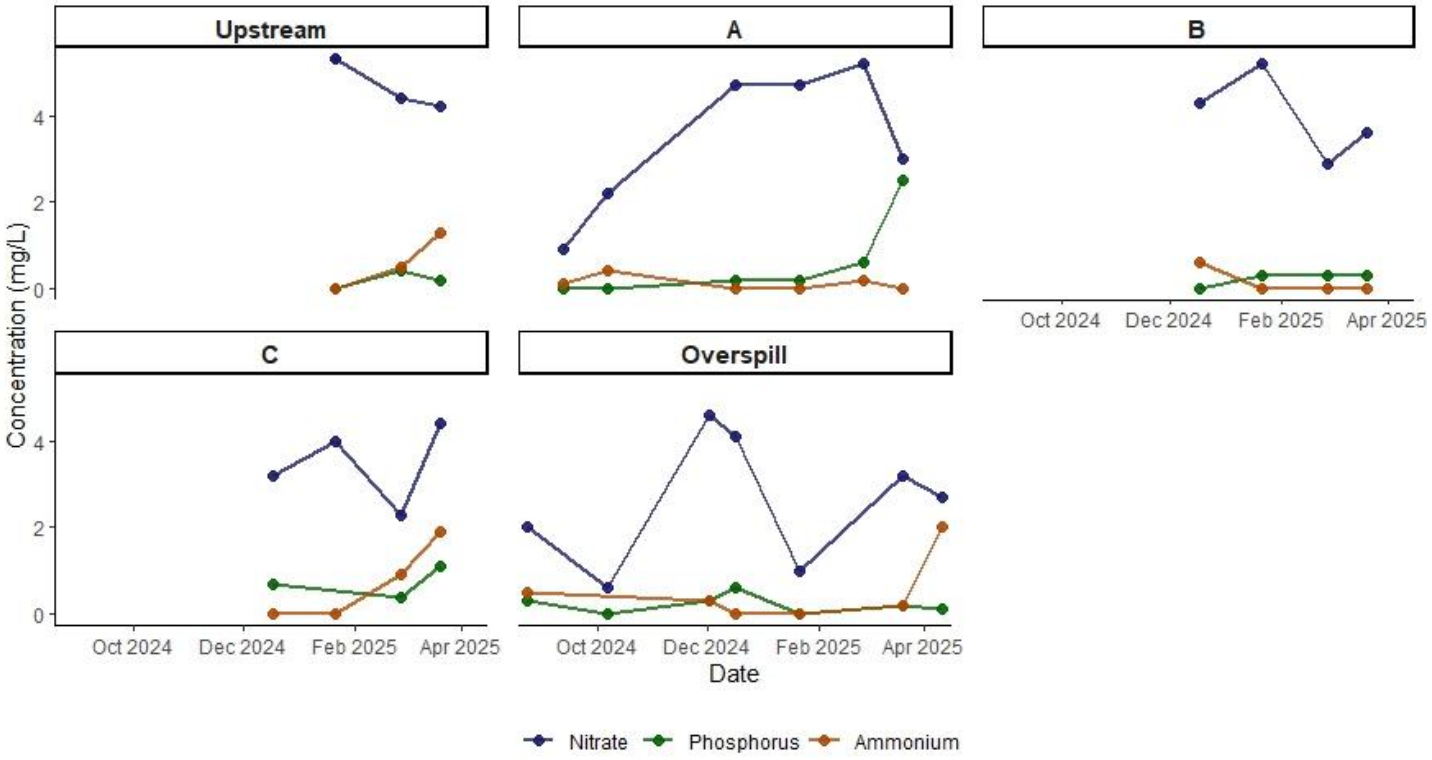


Figure 3.2: Natural England data showing changes in Nitrate, Phosphorus, and Ammonium concentrations in runoff at Cockle Ridge between August 2024 and May 2025. Concentrations were recorded at the Upstream, Wetland A, Wetland B, Wetland C, and Overspill sample locations. Data provided by Renfrey (pers. comm.).

At Kiln Point, the mean nitrate concentration across the three sample locations was 3.93 mg L^{-1} (*SD* = 2.15) with mean phosphorus and ammonium concentrations of 0.29 mg L^{-1} (*SD* = 0.16) and 0.37 mg L^{-1} (*SD* = 0.54) respectively *(Figure 3.3).* In March 2025, there is an anomalously low concentration of nitrate and phosphorus of 0 mg L^{-1} recorded at the Kiln Point inflow, which is not reflected by either of the two other sampling sites on the same date.

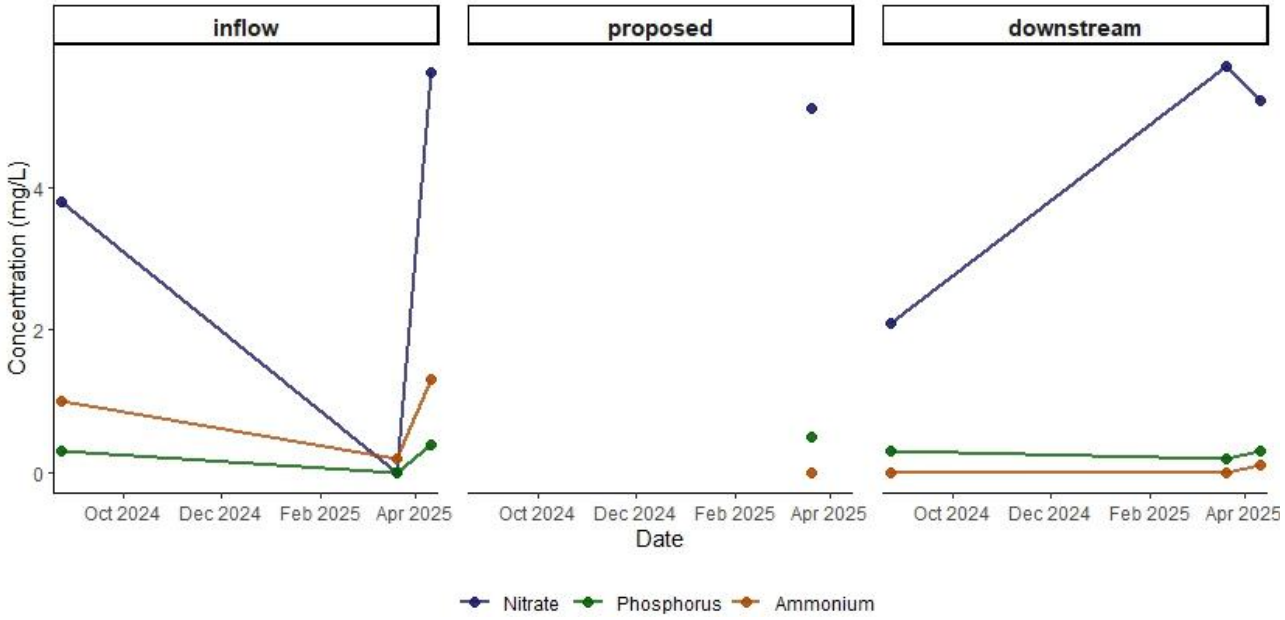


Figure 3.3: Natural England data showing changes in nitrate, phosphorus, and ammonium concentrations in runoff at the Kiln Point field drain between August 2024 and May 2025. Nutrients were recorded in 3 locations with the 'inflow' high up in field drain, 'proposed' at the location of the proposed 5-cell system, and the 'downstream' located at the outflow to the estuary. Data provided by Renfrey (pers. comm.).

3.2 Wetland Establishment Post-Construction

Following construction of the wetlands at Cockle Ridge in August 2024, the reeds that had been growing there previously were salvaged and added back into Wetland A near the water inflow from upstream. After nine months of wetland establishment, no surviving reeds were recorded at the Wetland A inflow in May 2025 (*Figure 3.4)*. However, some reeds were recorded on the edges of the wetland closer to the outflow to Wetland B *(Figure 3.5).*



Figure 3.4: Top of wetland A where water flow from upstream contributing stream. Left: (2024) salvaged reeds added back into the wetland from the previous reedbed after construction. Right: (2025) Top of wetland A after 9 months of wetland establishment. Left photo source: Renfrey (pers. comm.), right photo taken by Author.



Figure 3.5: Bottom of wetland A with a coir matting leaky dam into wetland B (centre) Left shows wetland during construction in August 2024 and Right shows bottom of wetland A after 9 months of wetland establishment. Some Phragmites australis *were recorded on the right side of the wetland, close to the outflow. Left photo source: Renfrey (pers. comm.), right photo taken by Author.*

Additionally, after nine months, Wetland A was primarily covered by a thick layer of algae across the whole pond surface, visible in *Figures 3.4 and 3.5*. By May 2025, the coir matting installed to the slope between Wetlands A and B was almost entirely colonised by *Nasturtium officinale (Figure 3.6).*



Figure 3.6: Coir matting installed to the slope between wetland A and B. Left: (2024) Photo taken after construction. Right: (2025) coir matting covered in vegetation. Left photo source: Renfrey (pers. comm.), right photo taken by Author.

During construction a boulder was installed in the inflow of Wetland C from the pipe culvert that runs underneath the public footpath. This baffle was installed to minimise direct waterflow towards the wetland outflow. In *Figure 3.7* this boulder is now only partially visible nine months after wetland establishment. Additionally, *Figure 3.4 – 3.7* illustrate vegetation cover has naturally regenerated following the groundworks associated with wetland construction.



Figure 3.7: Inflow pipe culvert from Wetland B into Wetland C underneath the public footpath. (Left) August 2024 following construction (Right) May 2025 after 9 months of wetland establishment. Left photo source: Renfrey (pers. comm.), right photo taken by Author.

As *Figure 3.4 – 3.7* show nearly all low-lying vegetation near the wetlands was disturbed during construction, the vegetation species observations recorded in *Table 2* are reflective of vegetation that has grown since the construction in August 2024. Photographs used for the vegetation observations are included in *Appendix I*. Vegetation observations were not recorded for the upstream sampling location as this is situated within a woodland and was not altered during the construction process. However, the downstream sampling location was included as species composition may change over time due to wetland construction. *Table 2* shows the most vegetation species were observed around Wetlands A and B and decreased towards the upper marsh with the fewest observed around the Overspill Pond and downstream drainage channel. Vegetation was recorded around each wetland cell. The dominant vegetation at Wetland A was the algae across the surface of the pond, at Wetland B it was *Nasturtium officinale,* at Wetland C and the Overspill Pond it was *Oenanthe crocata,* and downstream of the wetland cells it was *Phalaris arundinacea. Epilobium hirsutum* was observed at every sample location except from the sides of the cell system and downstream of the wetlands, closer to the upper marsh. Conversely, *Cochlearia Danica* was found at Wetland C and downstream of the cell system but was not found in any sites closer to the woodland.

Table 2: Vegetation species recorded at six sample location at Cockle Ridge. Species are ordered in alphabetical order and photographs which show the transition area between two wetlands are included in both rows. The dominant species for each wetland is highlighted in bold. Photographs used for the vegetation observations are included in Appendix I.

Sample location	Photo Numbers	Vegetation Species 1	Vegetation Species 2	Vegetation Species 3	Vegetation Species 4	Vegetation Species 5	Vegetation Species 6	Vegetation Species 7	Vegetation Species 8
Wetland A	1-5	*Epilobium hirsutum*	*Galium aparine*	**Algae**	*Phalaris arundinacea*	*Phragmites australis*	*Ranunculus repens*	*Rubus fruticosus*	*Urtica dioica*
Wetland B	5-7	*Anthriscus sylvestris*	*Epilobium hirsutum*	*Galium aparine*	*Lapsana communis*	***Nasturtium officinale***	*Oenanthe crocata*	*Pteridium aquilinum*	*Ranunculus repens*
Left & Right side of cell system	8-9	*Galium aparine*	*Oenanthe crocata*	***Pteridium aquilinum***	*Rumex obtusifolius*	*Sonchus asper*	*Taraxacum officinale*	*Urtica dioica*	
Wetland C	10-14	*Cochlearia danica*	*Epilobium hirsutum*	*Galium aparine*	*Nasturtium* officinale	***Oenanthe crocata***	*Phalaris arundinacea*	*Sonchus asper*	
Overspill pond	14-15	*Epilobium hirsutum*	***Oenanthe crocata***	*Phalaris arundinacea*					
Downstream drainage channel	16-18	*Cochlearia danica*	*Oenanthe crocata*	***Phalaris arundinacea***	*Urtica dioica*				

3.3 Cockle Ridge CW Performance

At Cockle Ridge, the analysed water samples showed total oxidised nitrogen (TON) was the nutrient in highest concentration at the first 3 sampling locations whereas ammonium was highest at the last 3 sampling location, closer to the downstream drainage channel (*Figure 3.8)*. The mean TON concentration at the upstream drain channel was 4.258 mg L^{-1} (*SD* = 0.491) whereas mean phosphorus and ammonium concentrations were 95.4% and 99.4% lower at 0.175 mg L^{-1} (*SD* = 0.038) and 0.024 mg L^{-1} (*SD* = 0.001) respectively. The lowest mean TON concentration recorded was 0.135 mg L^{-1} (*SD* = 0.055) at the Overspill Pond where mean phosphorus and ammonium concentrations were 0.058 mg L^{-1} (*SD* = 0.012) and 0.261 mg L^{-1} (*SD* = 0.130) respectively. In the downstream drainage channel, leading to the estuary the mean TON concentration was 0.243 mg L^{-1} (*SD* = 0.072), with mean phosphorus and ammonium concentrations of 0.058 mg L^{-1} (*SD* = 0.002) and 0.256 mg L^{-1} (*SD* = 0.065) respectively.

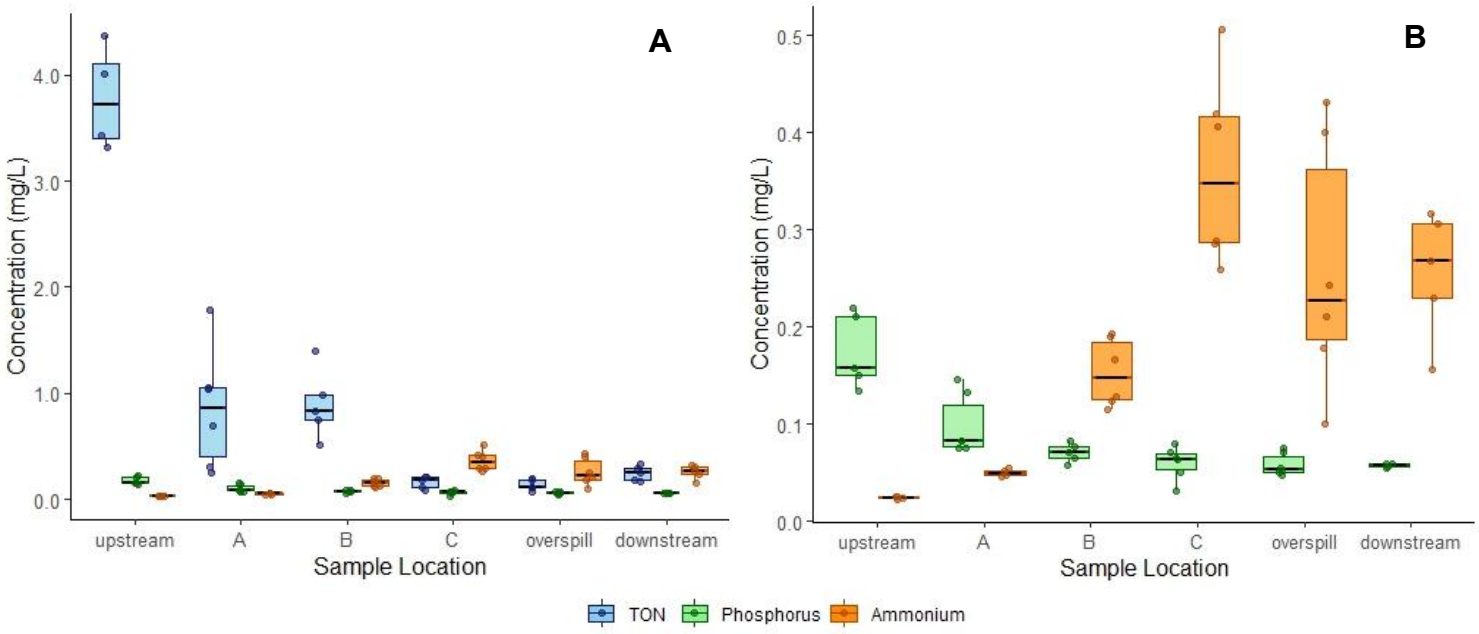


Figure 3.8: (A) The concentration of total oxidised nitrogen (TON), phosphorus, and ammonium in water taken at 6 sampling locations at Cockle Ridge, Budle Bay in May 2025 (B) Close-up of the phosphorus and ammonium concentrations on a second axis.

Compared to pre-intervention nutrient concentrations, both TON and phosphorus concentrations entering the estuary at Cockle Ridge have both decreased following the construction of the CW, whereas ammonium concentrations have increased (*Figure 3.9)*. TON concentrations entering the estuary decreased from 4.82 mg L^{-1} in 2017 to 0.243 mg L^{-1} in 2025, and phosphorus concentrations decreased from 0.078 mg L^{-1} to 0.069 mg L^{-1}

respectively. However, ammonium concentrations increased from 0.03 mg L^{-1} in 2017 to 0.256 mg L^{-1} in 2025.

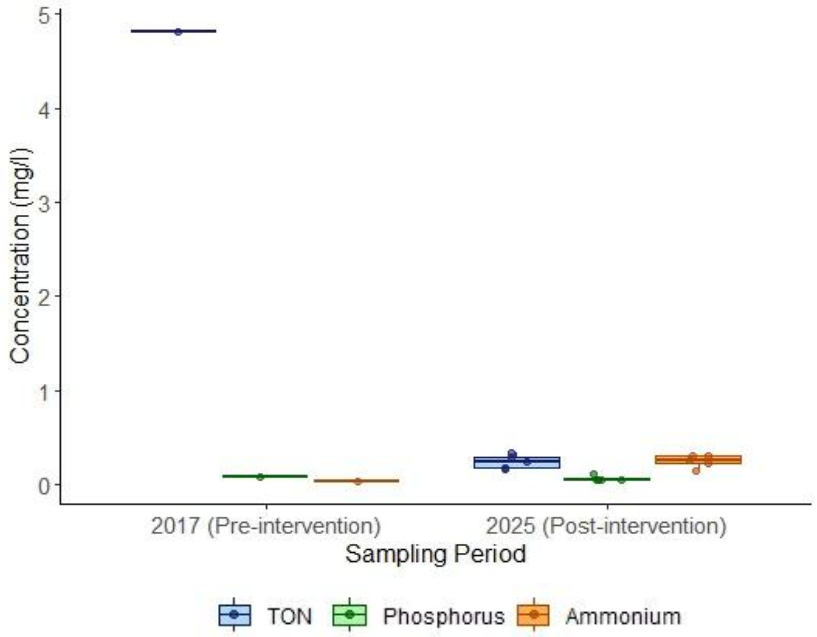


Figure 3.9: Comparing nutrient concentrations entering Budle Bay from Cockle Ridge in May 2017 (pre-intervention) and May 2025 (9-months post intervention). The 2017 data is provided by Environment Agency (Crook, pers. comm.), and the 2025 data is from the downstream drainage channel in this study. By using data from the same month in both years, this should avoid any issues of seasonality skewing results.

Both TON and phosphorus show a strong negative correlation with wetland cell number (ρ = -0.88, ρ = -0.84) (*Figure 3.10)*. The nutrients were both higher in mean concentration at Cell *0, 3.783* mg L^{-1} and 0.175 mg L^{-1} respectively, and decreased in concentration with increasing cell number, with mean concentrations of 0.135 mg L^{-1} and 0.058 mg L^{-1} respectively at Cell 4. TON concentrations decreased by 96.4% from Cell 0 to Cell 4, whereas phosphorus concentrations only decreased by 66.9%. Strong significant correlations were found between TON and wetland cell number ($P = 1.048 \times 10^{-8}$) and between phosphorus concentrations and wetland cell number ($P = 3.041 \times 10^{-8}$). These correlations were both significant at the 95% confidence interval. However, ammonium displays a positive correlation with wetland cell number *(*ρ = 0.79). Ammonium is found in low concentrations at Cell 0, a mean of 0.024 mg L^{-1}, and increases in mean concentration to a maximum of 0.361 mg L^{-1} at Cell 3, before decreasing to 0.261 mg L^{-1} at Cell 4. The correlation between ammonium and wetland cell number is significant at the 95% confidence interval ($P = 1.287 \times 10^{-6}$). Additionally, the inverse relationship *(*ρ = - 0.77) between TON and ammonium concentrations is significant at the 95% confidence interval ($P = 1.024 \times 10^{-6}$).

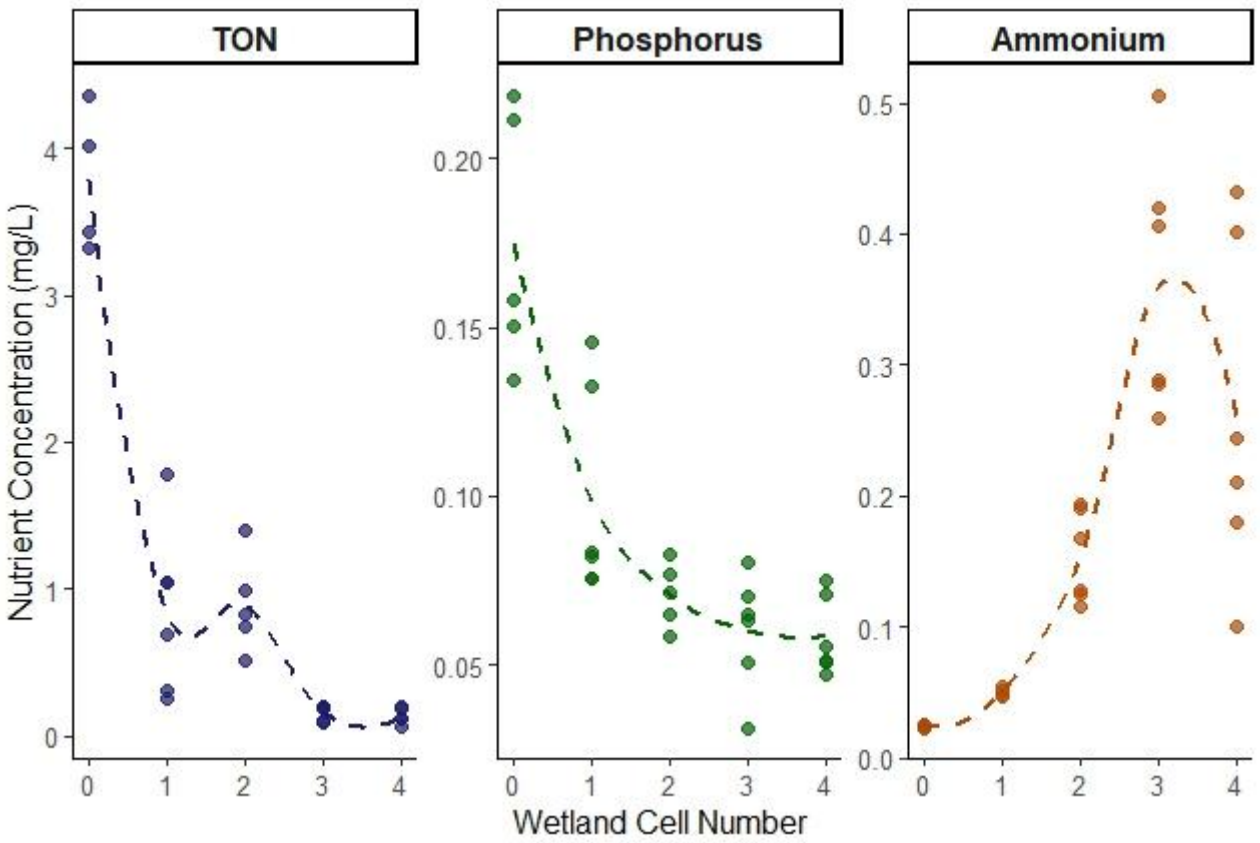


Figure 3.10: Relationship between total oxidised nitrogen (TON), phosphorus, and ammonium water concentrations, and wetland cell number where Upstream is cell 0 and Wetlands A-C and the Overspill Pond are cell numbers 1-4.

At Cockle Ridge, the 4-cell CW system saw a reduction in TON concentration of 96.43%, compared to the upstream input concentrations (*Figure 3.11).* The greatest reduction in TON was observed in the first wetland cell which accounted for 77.5% of the reduction. While there was an 19.5% increase in TON reduction ability between Cell 2 and 3 from 76.3% to 95.9%, the reduction ability plateaued in Cells 3 and 4 with an increase in TON reduction ability of 0.52% between the cells.

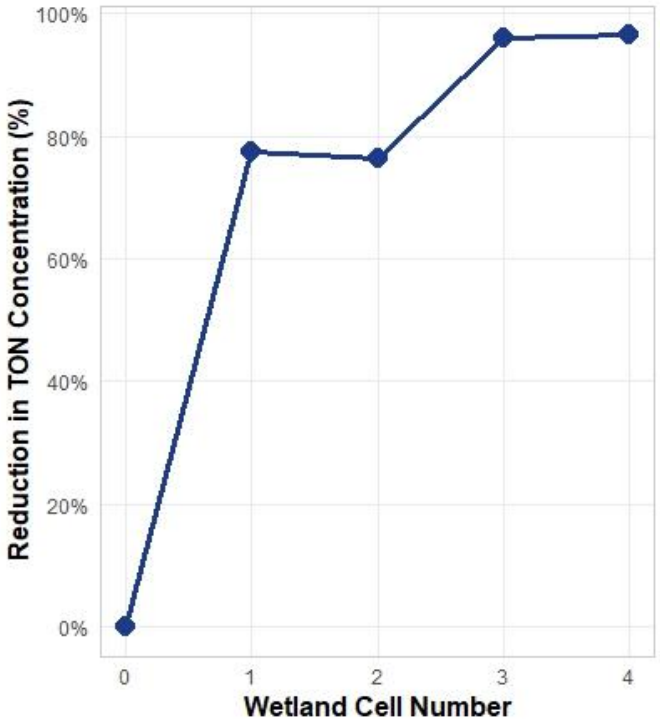


Figure 3.11: Recorded reduction (%) in total oxidised nitrogen across the 4 wetland cells at Cockle Ridge where Upstream is Cell 0 and Wetlands A-C and the Overspill Pond are cell numbers 1-4.

Within the wetland sediment, both TON and ammonium show a general positive trend, increasing in concentration through successive wetlands (*Figure 3.12)*. In Wetland A, the mean concentration of TON within the sediment was 0.207 mg kg^{-1} (*SD* = 0.052), increasing to 0.786 mg kg^{-1} (*SD* = 0.681) at the downstream sampling location. Ammonium was found in much greater concentrations, with a mean of 88.7 mg kg^{-1} (*SD* = 16.6) at Wetland A which increased to 103.6 mg kg^{-1} (*SD* = 95.3) at the downstream site.

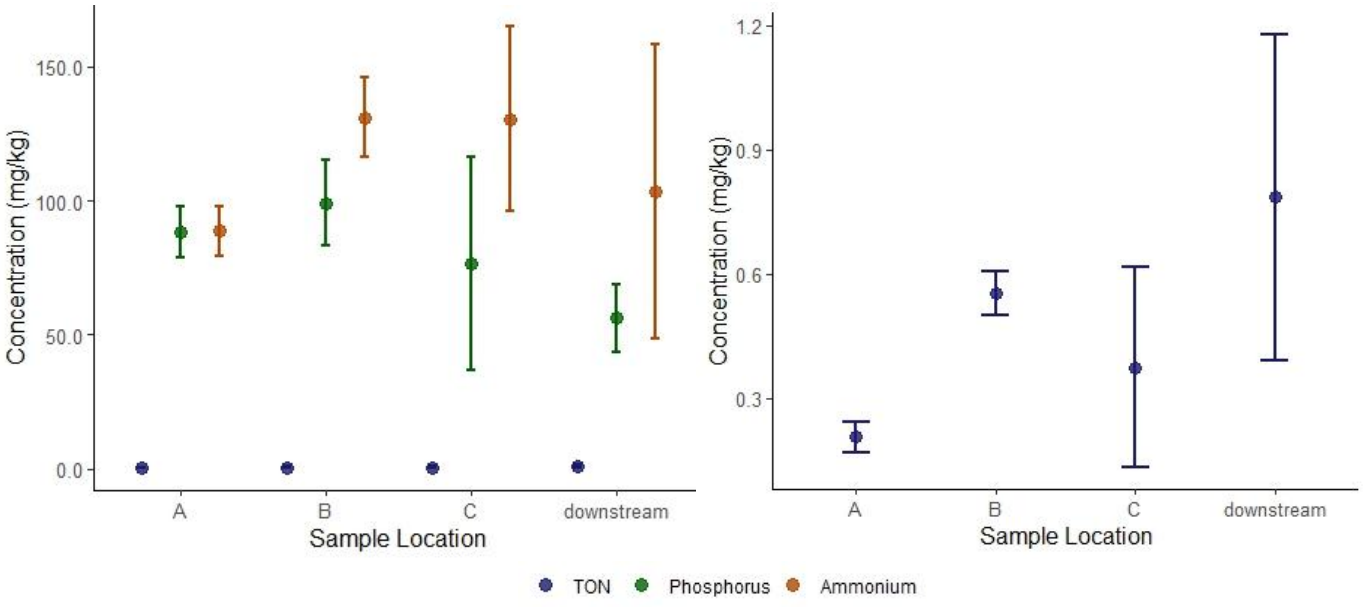


Figure 3.12: Total oxidised nitrogen (TON), phosphorus, and ammonium concentrations within sediment taken from the three constructed wetlands (A, B, C) and in the downstream drainage channel at Cockle Ridge leading into Budle Bay.

Conversely, phosphorus exhibited a negative trend, decreasing in concentration through successive wetlands (*Figure 3.12).* The mean concentration of phosphorus within the sediment at Wetland A was 88.4 mg kg^{-1} (*SD*= 16.3) which decreased to 56.2 mg kg^{-1} (*SD*= 22.0) at the downstream sampling location. All three nutrients displayed greater variability in sediment concentrations through successive wetlands, with increases in the standard deviations of TON, ammonium, and phosphorus of 92%, 83% and 26% respectively.

Sediment samples taken as reference samples from the left and right sides of the wetland cell system display higher concentrations of phosphorus and ammonium than TON (*Figure 3.13).* Additionally, sediment taken from the right side of the intervention was higher in all three nutrients than the sediment from the left side. The mean phosphorus concentrations were 78.9 mg kg^{-1} (*SD*= 13.1) on the left and 143.6 mg kg^{-1} (*SD* = 50.9) on the right, mean ammonium concentrations were 63.kg mg kg^{-1} (*SD* = 2.06*)* on the left and 70.6 mg kg^{-1} (*SD* = 0.583) on the right, and mean TON concentrations were 6.73 mg kg^{-1} (*SD*= 3.34) on the left and 32.0 mg kg^{-1} (*SD* = 1.56) on the right.

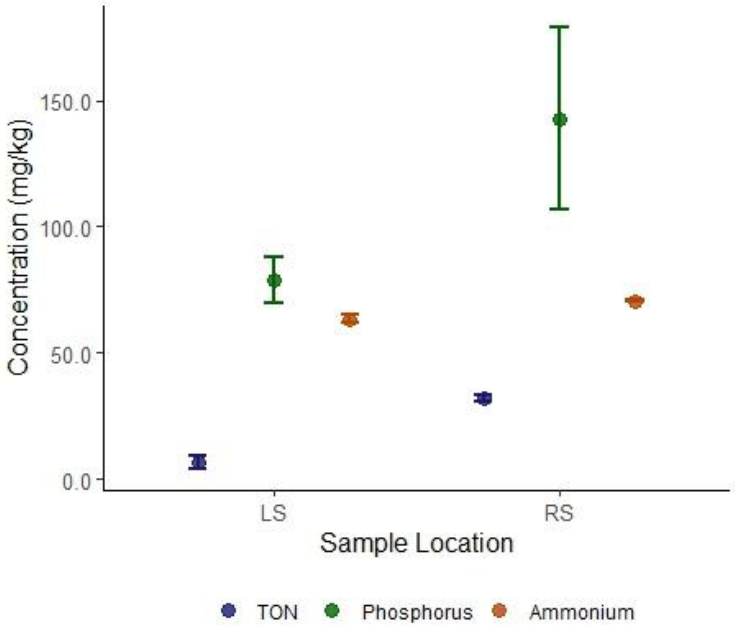


Figure 3.13: Total oxidised nitrogen, phosphorus, and ammonium concentrations within sediment taken from the sample location to the left side (LS) and right side (RS) of the Cockle Ridge wetland cell system.

3.4 Proposed Kiln Point CWs

At Kiln Point, TON is the nutrient in highest concentration within the water sampled (*Figure 3.14).* Mean TON concentrations were 6.99 mg L^{-1} (*SD* = 5.36) at the stream sampling location, where the wetland is proposed, and 6.85 mg L^{-1} (*SD* = 2.19*)* at the downstream outfall. Conversely, phosphorus and ammonium concentrations were substantially lower within the water samples; at the proposed wetland location zero phosphorus was recorded within the water, while mean ammonium concentrations were 99% lower than those of TON at 0.043 mg L^{-1} (*SD* = 5.36*)*. At the downstream outfall the mean phosphorus concentration was 0.004 mg L^{-1} (*SD* = 0.004) while the mean ammonium concentration was 0.027 mg L^{-1} (*SD* = 0.01*)*.

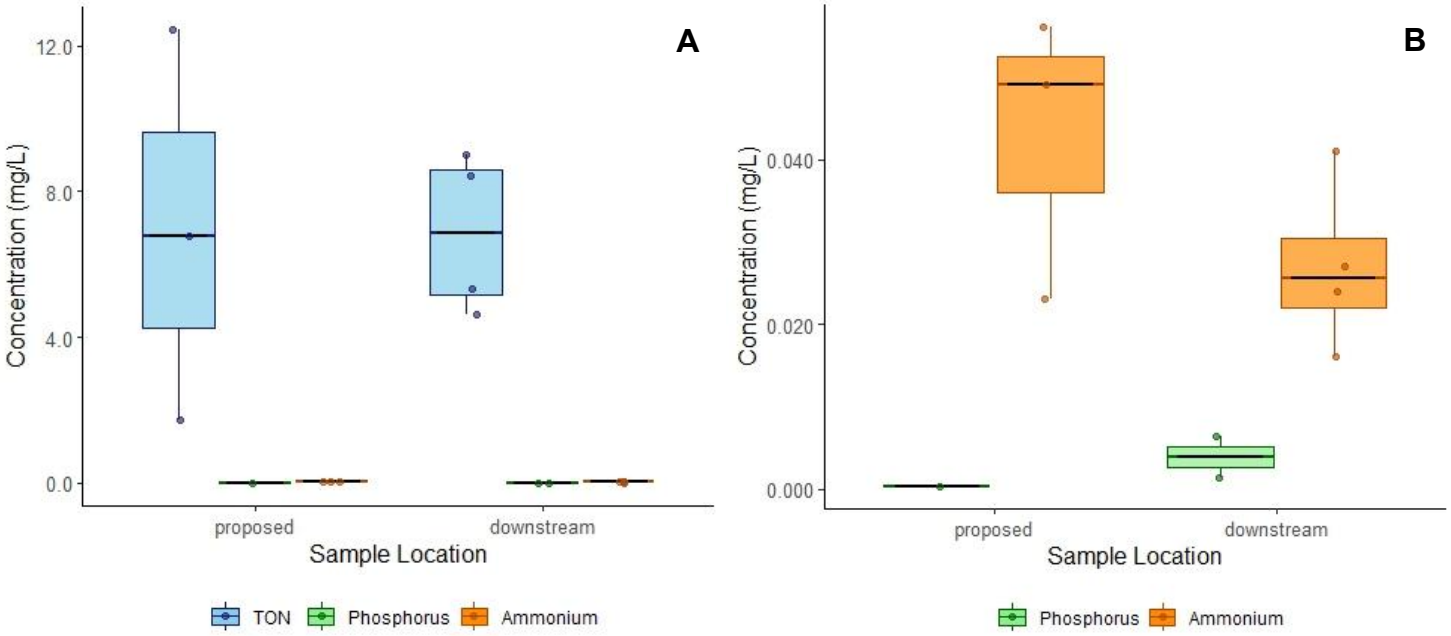


Figure 3.14: (A) Total oxidised Nitrogen, Phosphorus, and Ammonium concentrations in water at Kiln Point measured in the stream at the proposed wetland location and the downstream outfall into Budle Bay. (B) Close-up of Phosphorus and Ammonium on a second axis.

In comparison to Cockle Ridge, Kiln Point had higher mean concentrations of TON and ammonium in the stream draining into Budle Bay, whereas mean phosphorus concentrations were lower (*Figure 3.15).* The largest difference between the sites is the TON concentration. Mean stream TON concentrations were 4.26 mg L^{-1} at Cockle Ridge compared to 6.99 mg L^{-1} at Kiln Point. Ammonium concentrations were 0.02 mg L^{-1} higher at Kiln Point than Cockle Ridge, 0.043 mg L^{-1} and 0.025 mg L^{-1} respectively. Conversely,

phosphorus concentrations were substantially higher in the stream at Cockle Ridge, 0.175 mg L^{-1} compared to 0.0004 mg L^{-1} at Kiln Point.

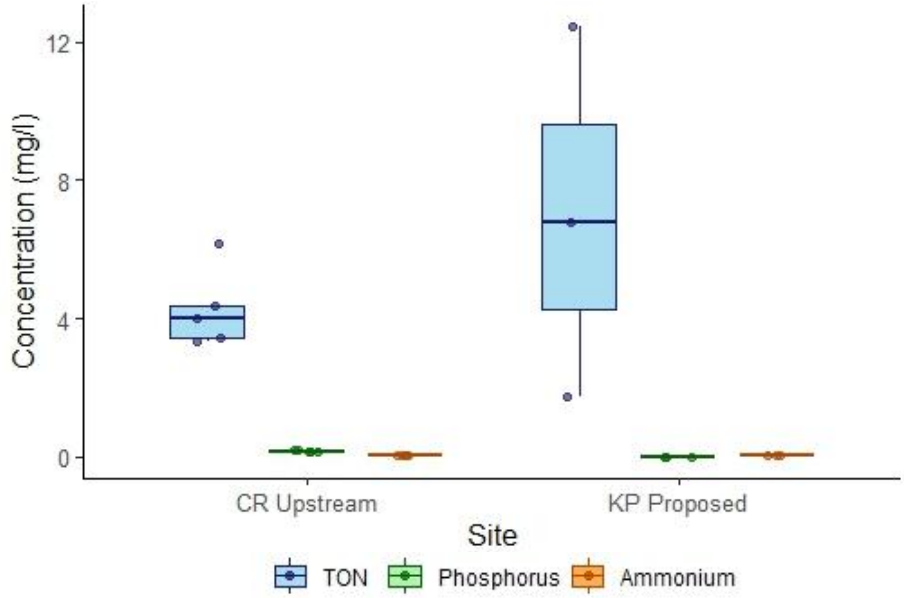


Figure 3.15: Comparing water nutrient concentrations in the field drains at Cockle Ridge and Kiln Point in 2025. This compares what present nutrient concentrations entering the estuary at both sites would be with no interventions present.

Similarly to Cockle Ridge, TON concentrations within the sediment at the Kiln Point stream were found to be extremely low, with a mean concentration of 2.41 mg L^{-1} (*SD* = 0.80) (*Figure 3.16).* However, at Kiln Point ammonium was the nutrient in highest concentration within the sediment. Ammonium had a mean concentration of 19.4 mg L^{-1} (*SD* = 5.38) whereas phosphorus only had a mean concentration of 3.11 mg L^{-1} (*SD* = 1.32), a concentration 96% lower than the lower concentration left side reference samples at Cockle Ridge.

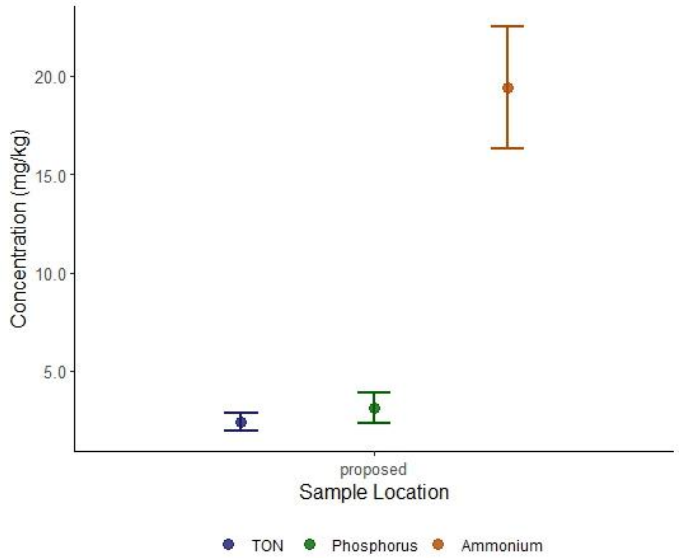


Figure 3.16: Sediment total oxidised nitrogen (TON), phosphorus and ammonium concentrations taken from the proposed wetland location in the stream at Kiln Point. No sediment could be collected from the downstream outfall due to the nature of its construction.

Using the trend in TON reduction observed at Cockle Ridge and by recognising a diminishing reduction in TON concentrations with increasing wetland cells (the reduction ability of a 5th wetland cell being 30% of that of the 4th cell), the TON reduction ability of a 5-cell system is predicted to be 96.59% compared to input concentrations (*Figure 3.17).*

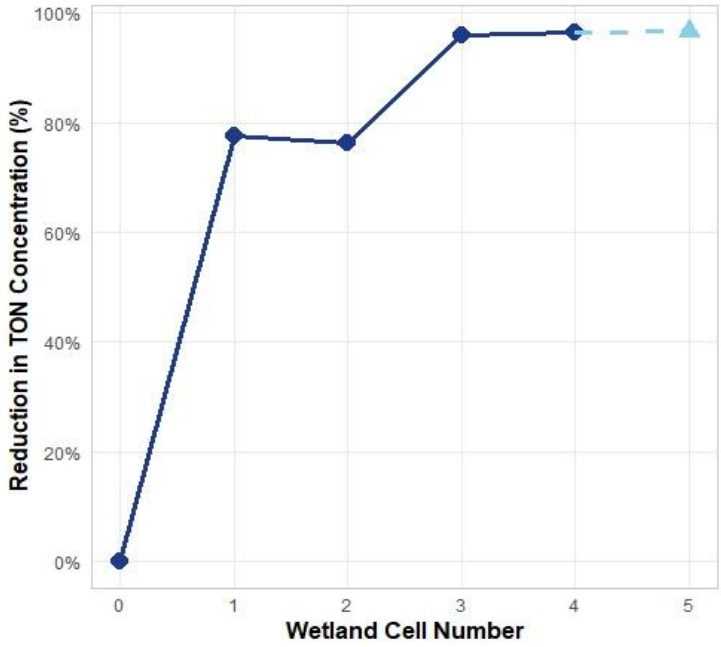


Figure 3.17: Recorded reduction (%) in Total Oxidised Nitrogen across the 4 wetland cells across Cockle Ridge and the predicted reduction (%) in TON if a fifth wetland was added calculated using simple trend analysis.

Using the simple trend analysis, in combination with the TON input (mg L^{-1}) recorded at the proposed wetland site at Kiln Point, and given the current plans are for a 5-cell system at Kiln Point, the predicted concentration of TON in the wetland outflow would be 0.24 mg L^{-1} when processed through a 5-cell wetland system, a 4% improvement upon a 4-cell system (0.25 mg L^{-1}) *(Table 3).*

Table 3: Potential outflow TON concentrations at Kiln Point given the reduction ability expected from a 4-cell system based off Cockle Ridge, and the reduction ability predicted for a 5-cell system by extrapolating from the Cockle Ridge data using simple trend analysis.

Wetland Location	Wetland cell number	Stream TON input (mg L^{-1})	TON reduction (%)	TON output from last wetland cell (mg L^{-1})
Cockle Ridge	4-cell *observed* TON reduction	4.26	96.43	*0.15*
Kiln Point	4-cell *expected* TON reduction	6.99	96.43	0.25
Kiln Point	5-cell *predicted* TON reduction	6.99	96.59	0.24

4 Discussion

4.1 Key Results Summary

- Total oxidised nitrogen (TON) is the nutrient found in highest concentration at both Cockle Ridge and Kiln Point and has remained so over time.
- There was substantial colonisation of vegetation, including both algae in Wetland A and the riparian vegetation surrounding the four cells after only nine-months post construction of the Cockle Ridge pilot intervention.
- The Cockle Ridge CW performed best removing TON from runoff, the nutrient in highest concentration in the upstream drainage channel and had an increasing negative correlation down through successive cells. Phosphorus followed this same pattern, while ammonium was found in highest concentration in Wetland C water, displaying a positive correlation through successive cells whilst also increasing in concentration compared to pre-intervention concentrations.
- The Kiln point field drain has higher TON water concentrations than the Cockle Ridge drainage channel, suggesting a need for a CW.
- As this study has demonstrated the success of the Cockle Ridge CW reducing TON concentrations by 96% it provides valuable evidence in favour of building CWs at Kiln Point with the likelihood of similar reduction in TON.

4.2 Historic Nutrient Concentrations in Budle Bay (Past)

Research Question 1: *What are the historic baseline nutrient concentrations at Cockle Ridge and Kiln Point, and how have conditions changed since the Cockle Ridge constructed wetland was installed in August 2024? (Past)*

Historic Nutrient Concentrations

The Environment Agency (EA) (2016/17) and Natural England (NE) (2024/25) monitoring data demonstrates TON, primarily in the form of nitrate, is the dominant nutrient pollutant entering Budle Bay, with mean concentrations ranging from 3.52-4.68 mg L^{-1} across both Cockle Ridge and Kiln Point outflows (*Figures 3.1-3.3)*. Phosphorus and ammonium concentrations remained comparatively low (<0.4 mg L^{-1}*),* confirming that these nutrients are not the primary concern for eutrophication management in the Budle Bay system.

The identification of nitrogen as the primary eutrophication concern is supported by previous research at Budle Bay, where greatest macroalgae abundance was found near areas of nitrogen-rich runoff (Warren Mill) compared to areas higher in phosphorus (Ross Low) (Arizona, 2022; Mill, 2009; Palmer, 2012). Therefore, LIFE WADER and NE should focus on nitrogen management as their primary remediation effort to minimise Budle Bay macroalgae.

However, the two monitoring locations showed different temporal patterns. Cockle Ridge, TON concentrations exhibited greater variability (*SD* = 1.88 in EA data, 1.41 in NE data) with seasonal patterns, indicating surface runoff nutrient inputs influenced by rainfall events and seasonal agricultural practices (Wang *et al.,* 2022) (*Figures 3.1 & 3.2)*. Conversely, the EA data at Kiln Point showed more consistent TON concentrations (*SD* = 0.288), without seasonal trends suggesting a groundwater-derived nutrient source (*Figure 3.1)***.** Future study is necessary to understand the nature of the nutrient source at Kiln Point.

Understanding of Kiln Point historic nutrient dynamics is complicated by anomalously low NE data in March 2025; nitrate and phosphorus concentrations of 0 mg L^{-1} at the field drain inflow *(Figure 3.3).* Similar anomalies were absent at Cockle Ridge on the same date, suggesting these low readings were site-specific rather than stemming from hydrological events (*Figure 3.2)*.

Additionally, limited recent Kiln Point data exists compared to Cockle Ridge, with only one 2025 data point at proposed Kiln Point CW location (*Figure 3.2 & 3.3).* More data was recorded in 2016/17 by the EA who conducted monthly monitoring. NE's more recent ongoing monitoring varies in frequency between sites and measures different nutrients (nitrates vs TON). This inconsistency in monitoring highlights the necessity for a coordinated, standardised monitoring methodology to evaluate the long-term performance of CWs.

Wetland Establishment

The CWs at Cockle Ridge exhibited substantial vegetation colonisation within nine-months, with all new species (except pre-existing *Phragmites australis*) establishing naturally. Clear ecological zonation emerged from woodland to upper marsh *(Table 2);* highest species diversity near Wetlands A and B, closest to the woodland, decreasing towards the upper marsh around the Overspill Pond and downstream drainage channel. This suggests proximity to the woodland facilitated more rapid and diverse colonisation, while increasing saline conditions near the upper marsh limited establishment to salt tolerant species like *Cochlearia danica,* a well-known halophyte (Fekete *et al.,* 2018). Most vegetation recorded

was riparian (*Epilobium hirsutum)* or emergent (*Phragmites australis* and *Nasturtium officinale)* (Bergey, 2024; Guittonny-Philippe *et al.,* 2015).

This substantial colonisation may have a notable impact on the CW's nutrient removal performance as vegetated wetlands remove more nitrates than unvegetated ones (Bouchard *et al.,* 2007; Lin *et al.,* 2002; Newman *et al.,* 2015). However, nitrogen attenuation is believed to be higher in emergent than submerged vegetation suggesting nutrient removal performance is more dependent on species identity than diversity (Yousaf *et al.,* 2021). Older and more established CWs are more effective at removing nutrients then newer ones. With more time to establish and emergent vegetation succession, the nutrient removal performance of these CWs can be expected to increase (Newman *et al.,* 2015; Robotham *et al.,* 2021).

The complete mortality of *P. australis* at Wetland A's deeper inflow, compared to their survival at the outflow, indicates that inflow conditions are unsuitable for these reeds. While high TON concentrations (4.26 mg L^{-1}) from the upstream channel could create toxic conditions, *P. australis* typically tolerate high nitrogen levels (Rohal *et al*., 2019). Instead, water depth may explain the mortality, as these reeds prefer shallow conditions and struggle in deeper waters (Song, 2021). Future CWs should consider water depth when positioning *P. australis* vegetation.

Extensive algae mats developed across Wetland A (*Figure 3.4 & 3.5),* which had the highest TON concentrations (*Figure 3.8).* This aligns with findings from similar CW studies where thick algae mats developed on wetland surfaces with high nutrient concentrations (Robotham *et al.,* 2021). Therefore, similarly to harmful algal blooms (HABs), algae growth on CWs can be used as biological quality indicators of high TON concentration (Gubelit, 2022).

Dense algae mats substantially reduce light penetration to the water column, limiting photosynthesis and submerged vegetation survival (Abonyi *et al.,* 2024; Caputo, 2022). Previous studies have also shown that anoxic conditions, created by algae mats, are highly injurious to emergent vegetation like *P. australis* (Armstrong and Armstrong, 2001). Therefore, future study should investigate CW dissolved oxygen concentrations and identify algae species to understand their impact on vegetation survival and nutrient removal.

Substantial sedimentation after nine-months, evidenced by the partially buried boulder baffle in Wetland C (*Figure 3.7),* indicates CW management is required. Sedimentation as one of

the most common problems in CWs and regular maintenance is required to avoid nutrient remobilisation and to extend the intervention's lifetime (Robotham *et al.,* 2021; Yousaf *et al.,* 2021). Dredging should be conducted in summer when water levels are lower, and the first wetland cell will require more frequent maintenance than following cells (Robotham *et al.,* 2021). Future monitoring should assess suspended sediment concentrations at both sites to develop tailored maintenance schedules.

Since this study was conducted during the May 2025 growing season when *P. australis* productivity and nutrient uptake are highest (Körner *et al*., 2023; Vymazal, 2007), the results represent peak vegetation nutrient removal. To fully assess vegetation's contribution to nutrient removal, this study should be repeated during autumn/winter when vegetation has died back.

As a single observation of young wetlands, this study provides insufficient data to correlate nutrient removal with vegetation presence. However, it establishes a valuable baseline for species composition nine-months post construction. Future studies should monitor how species change as wetlands become fully established (> 18 months) and assess how dominant species at each wetland *(Table 2)* change with changing nutrient concentrations (Newman *et al.,* 2015).

Therefore, to assess if seasonality, age, and vegetation presence impact CW nutrient removal performance, long-term consistent monitoring is needed which will remove point-in-time biases for example, grasses surrounding CWs were not flowering during fieldwork, possibly resulting in incorrect species identification.

Key take aways:

- *Nitrogen is the primary source of eutrophication in Budle Bay.*
- *A long-term extensive coordinated standardised monitoring methodology is required for a better understanding of nutrient origins and dynamics at Kiln Point and the impact of seasonality on nutrient dynamics.*
- *Future study should focus on investigating dissolved oxygen concentrations, suspended sediment concentrations and vegetation species composition over time.*

4.3 Cockle Ridge CW Performance in May 2025 (Present)

Research Question 2: *How effectively is the Cockle Ridge pilot constructed wetland reducing nutrient loads to Budle Bay nine months post- construction? (Present)*

- *Is treatment performance consistent across the 4-cell wetland system?*
- *What do the nutrient dynamics indicate about the wetland processes driving the observed treatment efficiency?*

CW Nutrient Reduction Performance

The Cockle Ridge CW demonstrated considerable effectiveness in reducing nutrient inflow to Budle Bay nine-months post construction compared to May 2017. The intervention achieved a 95% TON reduction, and an 11.6% phosphorus reduction compared to pre-intervention levels (*Figure 3.9)*. Additionally, within the intervention TON concentrations were reduced by 96.4% from the upstream drain channel (Cell 0) to the Overspill Pond (Cell 4) *(Figure 3.8).*

These results align with established CW performance benchmarks; the TON reduction aligns with 5 studies reporting CW nitrogen reductions of 90-99% (Abonyi *et al,* 2024*,* Barman *et al.,* 2024; Ferreira *et al.,* 2023; Hassan *et al.,* 2021; Parde *et al.,* 2020). This contrasts with earlier studies reporting only 72-83% nitrogen removal, indicating the Cockle Ridge pilot intervention performs at the upper end of expected effectiveness (Kim *et al.,* 2011; Mallin *et al.,* 2012). The 95% TON reduction also substantially exceeds the UK government's 2038 Environmental Target (Water) Regulations which aim for nutrient reductions of at least 40% compared to 2018 (DEFRA, 2023).

Low phosphorus reduction (11.6%) compared to pre-intervention aligns with other CW studies (Turcios *et al.,* 2021). Higher 2025 phosphorus inflow concentrations (0.17 mg L^{-1} *vs* 0.07 mg L^{-1} in 2017) may explain this low removal efficiency. However, phosphorus removal within the intervention was more substantial, decreasing by 66.9% from Cell 0 to Cell 4 *(Figure 3.8).* Therefore, more research is needed to understand if CWs will have the capacity to contribute to the government's 2038 40% reduction target.

TON and phosphorus concentrations were lowest at the Overspill Pond rather than downstream drainage channel, indicating additional runoff enters beyond the wetland outflow.

Ammonium concentrations increased by 88.3% compared to pre-intervention and by 90.8% from Cell 0 to Cell 4 *(Figure 3.8 & 3.9),* consistent with NE's 2024/25 data *(Figure 3.2).* Other studies reported similar increases, with Omidinia-Anarkoli and Shayannejad (2024) explaining that low baseline ammonium inputs can disrupt microbial processes, leading to elevated ammonium in the treated water. However, ammonium concentrations showed substantial standard deviations, indicating high data variability that may affect the reliability of these percentage increases at Cockle Ridge.

The low phosphorus removal and ammonium increase, while concerning, should be contextualised within overall intervention nutrient dynamics. TON dominated nutrient inputs, with phosphorus and ammonium concentrations being 95.4% and 99.4% lower respectively than TON in the upstream drain channel *(Figure 3.8).* Therefore, TON is the primary water quality challenge, with phosphorus and ammonium constituting minor pollution sources. To address increased ammonium concentrations, enhancing species richness (especially in Cells 3-4 where lower species richness was recorded) and introducing ammonium-tolerant duckweed species, like *Landoltia puncata 0202,* are measures proven to reduce ammonium concentrations (Ge *et al.,* 2015; Li *et al.,* 2025; Tian *et al.,* 2021).

Multi-cell Treatment Consistency

Treatment performance across the 4-cell intervention demonstrated both effectiveness and variability depending on the nutrient. As TON was identified as the primary eutrophication threat to Budle Bay, the individual wetland cell dynamics were examined in detail (*Figure 3.12).* TON concentrations decreased 96.4% from Cell 0 to Cell 4, substantially outperforming phosphorus (66.9% reduction) and ammonium (90.8% increase). Therefore, this 4-cell pilot intervention demonstrated superior capability for TON removal (*Figure 3.10)*.

Treatment performance was unevenly distributed with Wetland A (Cell 1) accounting for 77.5% of achieved TON reduction (*Figure 3.12)*. Performance plateaued in subsequent cells with Cell 4 contributing only 0.52% additional improvement compared to a 3-cell system, questioning the cost-effectiveness of a 4-cell design when similar treatment is achieved by a more compact 3-cell intervention. However, increased hydraulic residence time (HRT) rather than cell number drives the increased removal efficiency of multi-cell systems (Ferreira *et al.,* 2023; Hassan *et al.,*2021). Therefore, future research should compare the removal efficiency of systems with fewer cells of greater areas versus more cells of smaller area to optimise intervention design recommendations.

Total Oxidised Nitrogen

The substantial TON reduction across the Cockle Ridge pilot intervention may be driven by microbial processes rather than plant uptake or sediment adsorption. While sediment TON concentrations increased across the intervention indicating some soil adsorption, concentrations were minimal compared to phosphorus and ammonium, therefore unlikely to account for the substantial TON reduction (*Figure 3.11).* While microbial nitrification and denitrification are the most common nitrogen removal mechanisms in wetlands, nitrification is unlikely to be the primary process at Cockle Ridge since this converts ammonium to TON, opposite to observed patterns (Robotham *et al.,* 2021). Nitrification also requires aerobic conditions which algae presence in Wetland A limit (Jamieson *et al.,* 2003).

Evidence suggests denitrification (microbial conversion of TON to gaseous nitrogen) is the dominant removal mechanism. This process increases under anoxic or low-oxygen conditions and is therefore enhanced by algae mats which create oxygen-depleted environments (Cheng *et al.,* 2016; Cheng *et al.,* 2022; Lin *et al.,* 2002; Robotham *et al.,* 2021; Yousaf *et al.,* 2021). This may explain Wetland A's treatment dominance.

Further research should investigate how dissolved oxygen concentrations correlate with nutrient concentrations. Since denitrification increases with temperature, the May study period could represent near-optimal TON removal conditions (Mallin *et al.*, 2012; Yousaf *et al.*, 2021). This temperature dependency suggests CWs may show seasonal variation, with peak efficiency during warmer months, meaning future studies across multiple seasons are required.

Ammonium

The 90.8% increase in water ammonium concentrations across the 4-cell intervention may be driven by dissimilatory nitrate reduction to ammonium (DNRA). Sediment ammonium concentrations increase through Wetlands A to C and are nearly double reference side sediment levels (*Figure 3.11 & 3.12)*. This indicates substantial ammonium settling from water and adsorbing to sediment. However, with low concentrations present in inflow water *(Figure 3.8),* ammonium must be created through another process.

DNRA converts TON into ammonium, explaining both increasing ammonium and decreasing TON concentrations (Wang *et al.,* 2024). As DNRA favours higher oxygen environments, this explains why ammonium increases substantially in Wetlands B and C compared to A, peaking in Wetland C (*Figure 3.8)* (Li *et al.,* 2020; Quick *et al.,* 2019; Wang *et al.,* 2024). Algae in Wetland A create anoxic conditions preventing DNRA while Wetlands B and C, without algae mats, have sufficient oxygen for rapid DNRA.

Water and sediment ammonium concentrations decrease at the Overspill Pond and downstream drainage channel (*Figure 3.8, 3.10, 3.11).* While sufficient oxygen exists for DNRA, TON concentrations strongly control this process and become limiting for TON to ammonium conversion (Deng *et al.,* 2015; Kraft *et al.,* 2014; Wang *et al.,* 2024).

Microbial processes are impacted by nutrient concentrations; environments with low ammonium inputs experience low nitrification (ammonium conversion to TON) (Jamieson *et al.,* 2003). Therefore, removal rates are low in the first wetland cells due to the low ammonium input at Cockle Ridge, and the algae created anoxic conditions (*Figure 3.8)* (Vymazal and Kropfelova, 2015). However, at the Overspill Pond and downstream drainage channel, ammonium concentrations are higher meaning sufficient concentrations may exist for nitrification, causing the observed ammonium decrease (*Figure 3.10)* (Omidinia-Anarkoli and Shayannejad, 2024; Vymazal, 2007). The slight TON increase in the downstream drainage channel compared to the Overspill Pond provides evidence of nitrification.

The nutrient dynamics and microbial processes occurring in each wetland cell are summarised in *Figure 4.1.*

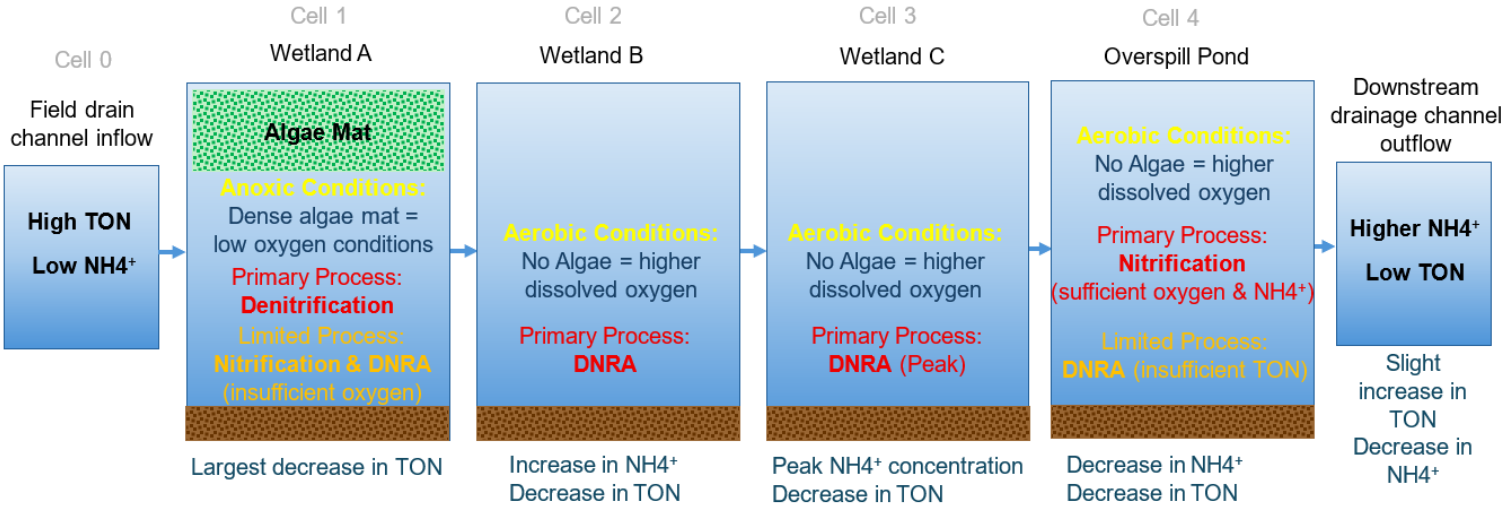


Figure 4.1: Conceptual diagram of the microbial processes thought to be occurring at the Cockle Ridge pilot intervention. Processes involved in the conversion between TON and ammonium nitrogen species, the conditions they require, and conditions which limit them are illustrated.

Future research should assess if the correlation between increasing dissolved oxygen and increasing ammonium concentration is significant, particularly since this study already found significant correlation between decreasing TON and increasing ammonium. Given that literature debates whether DNRA is anaerobic (Zhu *et al.,* 2023) or aerobic (Li *et al.,* 2020), oxygen monitoring of CWs is needed to help explain nutrient removal processes (Robotham *et al.,* 2021; Wang *et al.,* 2024).

Phosphorus

Phosphorus concentrations in both water and sediment decreased throughout the intervention (*Figure 3.8 & 3.11)* with sediment concentrations similar to reference levels with high variability (*Figure 3.11 & 3.12).* This suggests removal occurs primarily through plant uptake rather than soil adsorption (Kim *et al.,* 2011; Mallin *et al.,* 2012). However, phosphorus concentrations plateau between Wetland C and the downstream channel (*Figure 3.8),* indicating limited removal capacity in later wetland cells. This may indicate vegetation involved in phosphorus uptake is only present in the first two cells, with vegetation harvesting required in later cells to maintain treatment efficiency (Ennabili and Radoux, 2021).

This contradicts previous studies identifying sedimentation and soil matrix adsorption as primary phosphorus removal mechanisms in CWs (Maucieri *et al.,* 2020). Further phosphorus nutrient dynamics study is required and should include monitoring impacts on phosphorus removal efficiency due to plant uptake changes associated with seasonality.

Implications

The Cockle Ridge pilot intervention successfully addresses the primary eutrophication threat to Budle Bay as TON represents the highest pollutant load and CWs effectively remove TON. Therefore, results strongly support implementing CWs at Kiln Point, another TON dominated catchment and throughout the wider Lindisfarne National Nature Reserve (LNNR).

Treatment efficiency is concentrated in the first wetland cell, with diminishing returns beyond the third cell. Given most TON reduction occurs in the first cell, even single-cell CWs could provide substantial water quality improvements suggesting that where space limits multi-cell interventions, implementing a simplified single-cell design would still deliver substantial environmental benefits.

The 90.8% ammonium increase across the intervention represents a potential concern that could threaten the LNNR if left unchecked. Excessive ammonium causes oligotrophication, a process where phytoplankton growth is repressed rather than stimulated as in eutrophication, so the estuary is nutrient rich, but low in productivity (Helleman *et al.,* 2017). Microbial processes are also disturbed and an anoxic environment created, further stressing the ecosystem (Wilkerson and Dugdale, 2016). However, ammonium concentrations remain negligible compared to TON inputs and well below oligotrophication thresholds, making this increase acceptable given the substantial TON reduction achieved.

Future monitoring should prioritise tracking ammonium and phosphorus concentrations as the system matures, particularly whether ammonium concentrations continue increasing relative to inputs and if phosphorus levels continue rising above 2017 baselines. Long-term monitoring of all three nutrients will be essential to understand evolving nutrient dynamics, seasonal changes, the balance between TON removal and ammonium production, and for the development of strategies to address continued phosphorus increases across the catchment. Monitoring dissolved oxygen concentrations will improve understanding of nutrient removal processes and help balance conditions for TON removal (denitrification and DNRA) without producing excessive ammonium.

Key take aways:

- *Results strongly support implementing CWs at Kiln Point, another TON dominated catchment.*
- *The first wetland cell provides the majority of observed treatment.*
- *Wetlands primarily treat TON, through denitrification, DNRA, and soil adsorption.*
- *There is limited phosphorus treatment, requiring further investigation.*
- *Ammonium is being produced – in much lower quantities than TON but needs careful monitoring.*

4.4 CW Potential at Kiln Point (Future)

Research Question 3: *Based on comparative nutrient loading and observed treatment performance at Cockle Ridge, and current site conditions at Kiln Point, what is the predicted effectiveness of the proposed Kiln Point constructed wetland, and what are the recommendations for its implementation and monitoring strategy? (Future)*

Current Conditions

Monitoring at Kiln Point revealed substantially higher water TON concentrations than Cockle Ridge, with mean field drain concentrations of 6.99 mg L^{-1} (*SD* = 5.36), 3.06 mg L^{-1} higher than NE's monitoring and 2.73 mg L^{-1} higher than Cockle Ridge's upstream drain channel (*Figure 3.14 & 3.15)*. This indicates Kiln Point poses a greater eutrophication threat to Budle Bay than Cockle Ridge. While some of the difference from NE monitoring data may be attributed to NE only tested for nitrate rather than TON, the high standard deviation of the few Kiln Point samples suggests substantial concentration variability. This highlights the need for more comprehensive sampling at this site to better understand the temporal patterns and scale of the nutrient issue before developing any CWs.

The field drain at Kiln Point also exhibited higher ammonium concentrations than Cockle Ridge (*Figure 3.8 & 3.14).* Since the Cockle Ridge CWs increased water ammonium concentrations, intervention construction at Kiln Point may further elevate ammonium concentrations flowing into Budle Bay. These ammonium concentrations remain negligible compared to TON concentrations and show high data variability (*Figure 3.14).*

Sediment analysis showed similar patterns to Cockle Ridge (low TON, higher phosphorus, and highest ammonium) but with notably lower overall nutrient concentrations, potentially reflecting the lack of intervention (*Figure 3.16)*. The free-flowing field drain lacks sufficient HRT for nutrient settling and soil adsorption (Ferreira *et al.,* 2023; Hassan *et al.,*2021). To confirm this, reference sediment samples should be collected from undisturbed land surrounding the field drain against which these sediment nutrient concentrations can be compared.

The contrasting nutrient input characteristics between sites have important management implications. While CWs may effectively treat Cockle Ridge's variable, runoff-driven inputs through seasonal uptake mechanisms, Kiln Point's consistent groundwater-derived inputs present different challenges. If wetland treatment efficiency decreases in winter when plant

uptake is reduced, continuous groundwater nutrient supply could maintain elevated nutrient flow to the estuary year-round. Therefore, confirming nutrient sources at Kiln Point and investigating seasonal CW effectiveness are critical knowledge gaps requiring further study.

Recommendations for CW installation at Kiln Point

As a TON dominated site, Kiln Point should respond well to CWs based on Cockle Ridge evidence. A 5-cell vegetated intervention could achieve a 96.59% reduction in TON in summer (*Figure 3.17)*, lowering concentrations from the current 6.99 mg L^{-1} to 0.25 mg L^{-1} (*Table 3).* However, the fifth cell may provide minimal additional benefit over a 4-cell system already operating at > 95% efficiency.

Assuming the proposed CW at Kiln Point operates similarly to Cockle Ridge, a rise in ammonium concentrations entering the estuary is expected. The negative correlation between TON and phosphorus with wetland cell number, and positive correlation with ammonium, suggests these same patterns could emerge at Kiln Point. Once wetland cells are constructed, a rise in sediment nutrient concentrations is expected as increased HRT allows for nutrient settling and adsorption to the soil matrix (Ferreira *et al.,* 2023; Hassan *et al.,* 2021).

However, a study by Rodriguez-Dominguez *et al.* (2020) found CW treat performance reduces with increasing nutrient concentrations. As Kiln Point has higher TON concentrations than Cockle Ridge, there is potential that CW treatment may be less effective than anticipated. To mitigate the higher TON concentration at Kiln Point, as well as the additional wetland cell, more regular maintenance and deepening the cells could help enhance the nutrient removal capacity (Deng *et al*., 2024).

Comprehensive pre-intervention monitoring is required to establish accurate baseline data for Kiln Point to help address the high standard deviation currently observed. Additionally, the nutrient input source should be identified, and reference sediment samples collected to determine whether observed sediment differences are due to wetland influences or natural soil variations in the area.

Wider Implications

The Cockle Ridge pilot multi-cell intervention achieved a 95% reduction in TON concentrations compared to pre-intervention levels, with the first cell alone providing 77.5%

of the reduction. This demonstrates CWs effectively remove TON, the primary eutrophication threat in Budle Bay, protecting the LNNR from HABs and associated ecological damage if all nutrient sources are addressed.

As agriculture is the primary nutrient source at Cockle Ridge, as well as implementing CWs as a delivery-control method, local landowners should be encouraged to follow good agricultural practices such as avoiding fertiliser application before heavy rainfall and ploughing fields to increase soil infiltration (Haygarth *et al.,* 2009; Holden *et al.,* 2017; Preedy *et al.,* 2001).

Given their success, CWs represent a cost-effective NbS for TON-dominated catchments facing eutrophication risks; Cockle Ridge cost ~ £5000. In early 2025, a second round of CW site selection began at the LNNR; 7 sites were identified as priorities for further CW implementation to enhance water quality however, only 3 sites are under active consideration for multi-cell interventions due to land holding and budget constraints (Renfrey, pers. comm.).

Since single cell wetlands achieved a 77.5% reduction in TON with lesser land requirements and costs, Natural England should consider this design approach when selecting other sites across the LNNR. Multiple smaller interventions would reduce eutrophication risk to both Budle Bay and Holy Island Bay while providing co-benefits like increased biodiversity (Abonyi *et al.* 2024; Barman *et al.,* 2024; Cheng *et al*., 2020). Beyond the LNNR boundaries, several ‘high priority’ Water Priority Areas (WPA) pose additional threats, particularly the Till (Tweed) catchment with an outflow at Berwick-upon-Tweed, just north of the reserve (Natural England, 2016). This catchment, dominated by TON and phosphorus pollution, could damage the surrounding environment with potential knock-on impacts to the LNNR itself, making CWs a valuable intervention for this area as well.

CW implementation potential extends beyond the LNNR and estuaries to address the UK's most pressing eutrophication challenge: Lough Neagh, Northern Ireland, a prominent inland waterbody (Jordan *et al.,* 2024). Like Lindisfarne, eutrophication is driven primarily by agricultural nitrogen runoff with some contribution from sewage overflow. Current improvement strategies focus on enhanced agricultural practices and WwTW upgrades but notably omit NbSs like CWs (Cave and Allen, 2023).

Lough Neagh has experienced several blue-green algae blooms, most notably in summer 2023, compromising local fisheries and drinking water supplies while producing dangerous

cyanotoxins *(Figure 4.2)* (European Space Agency, 2023b). As an SSSI, SPA, and Ramsar designated site like Lindisfarne, this large-scale eutrophication has severe implications for this environmentally important location (DAERA, 2025). The lake currently holds 'poor ecological status' under the EU Water Framework Directive.



Figure 4.2: Blue-green algae bloom in Lough Neagh, Northern Ireland in September 2023. The algae (green swirls) are most obvious on the eastern edge of the lake. Satellite imagery taken from European Space Agency (2023b).

Given that eight rivers and numerous tributaries channel agricultural runoff directly into Lough Neagh, and CW's proven ability to handle TON pollution, strategic intervention across these many waterways could provide substantial water quality improvements (Cave and Allen, 2023). This represents an ideal case for multiple small interventions across many streams creating a combined large-scale reduction in eutrophication.

4.5 Limitations

This study identified three key limitations:

1. Historical data was more limited at Kiln Point than Cockle Ridge, with a correspondingly reduced sampling effort during this study. The high standard deviation in TON results at Kiln Point suggests these samples should be repeated to

better understand present nutrient concentrations. Since TON results at Cockle Ridge showed lower standard deviation, the variability is location-dependent rather than methodological.

2. A high standard deviation in ammonium results at both locations may indicate contamination during sample processing. While sediment blank contamination from extraction solution was identified and corrected (*section 2.5.2*), high variability persisted in ammonium water samples, suggesting issues with either the ammonium specific reagent or the auto-analyser processing. Despite the high variability in results, this does not negate the general trends observed.
3. The results only reflect spring/summer treatment performance in CWs due to mid-May fieldwork timing. Therefore, a comparison study of autumn/winter treatment performance in CWs is needed to understand seasonal variations in treatment efficiency, as these results cannot be assumed to be representative of year-round CW performance.

4.6 Monitoring, Research, and Recommendations (Future)

Monitoring

Future monitoring conducted by NE, EA or research projects should follow a coordinated, standardised methodology to enable long-term understanding of CW interventions. This methodology must be clearly communicated to ensure consistent sampling frequencies and locations. *Table 4* outlines essential parameters for future monitoring based on this study's findings, with detailed frequency requirements and rationales. A coordinated approach would improve comparability between agencies and enhance understanding of CW effectiveness as the interventions mature.

Pre-construction baseline monitoring should begin at Kiln Point several months before construction. Since this study demonstrated successful collection and freezing of water and sediment samples, samples following the standardised methodology could be taken at minimum one month before construction, collecting the seven parameters from *Table 4* and storing them frozen for future analysis.

Table 4: Parameters to be included in future monitoring to better understand the functioning of constructed wetlands. The recommended frequency and rationale of including each parameter is detailed.

Parameter	Frequency	Rationale
Total Oxidised Nitrogen (TON)	Monthly	Primary pollutant: standardise method used across agencies
Ammonium	Monthly	Monitoring increases within wetland and outflow to estuary
Phosphorus	Monthly	Understand seasonal plant uptake variations
Dissolved Oxygen	Monthly	Understanding denitrification/ DNRA processes, algae impact, and vegetation survival
Suspended Sediment Concentration	Monthly	Maintenance scheduling and treatment efficiency
Vegetation Species Composition	Seasonal	Treatment performance correlation
pH	Monthly	Historical and present data available but underutilised
Temperature	Monthly	Understanding seasonal variations

Research Priorities

Key future research priorities for expanding CW knowledge, identified throughout this study, are summarised in *Table 5.* While this research analysed spatial nutrient changes across an intervention, the primary priority for future work is investigating temporal changes, particularly seasonal impacts changes and wetland maturation effects on treatment efficiency.

Future research should also incorporate pH analysis. Although pH data was collected during this study, analysis was beyond this study's scope due to the extensive analytical requirements. This data is included in *Appendix II* for future use, as pH is known to influence HAB occurrence (Carvahlho *et al.,* 2013).

Table 5: Key future research priorities to expand knowledge on constructed wetland functioning.

Research Priority	Method	Expected Outcome
Seasonal impact	Quarterly sampling across seasons	Understand how treatment efficiency and nutrient input varies with season
Wetland maturation	Compare performance of this study with data from >18-month of establishment	Understand how wetland performance changes with maturation.
Rainfall response	Undertake sampling after high rainfall and drought periods	Understand external forcing impacts on treatment performance and nutrient input and remobilisation of nutrient adsorbed to soil matrix
pH impact	Utilise historical and present pH data and continue monitoring in the future	Understand other possible controlling factors of treatment efficiency
Kiln Point nutrient source	Investigate the source of the Kiln Point field drain and conduct groundwater sampling	Understand the nature of nutrient inputs at Kiln Point
Dissolved oxygen	Sampling of dissolved oxygen concentration variations across intervention	Understanding nutrient removal processes and correlation with algae presence

Next Steps & Design Recommendations

Priority next steps for NE based on this study's evidence are detailed in *Table 6.* Immediate actions include developing a standardised monitoring protocol, conducting pre-construction baseline monitoring at Kiln Point, and constructing the Kiln Point 5-cell intervention. Within 6-18 months, the Cockle Ridge Overspill Pond should be formalised into a permanent fourth cell, pilot intervention maintenance undertaken, and more TON-dominated catchments identified across the LNNR. Long-term (>18-months) maintenance schedule of each CW should be optimised based on monitoring, and ammonium concentration trends monitored to prevent oligotrophication.

Table 6: Programme of next steps to be taken by Natural England (NE) to ensure coordinated sampling methodology is followed and aid the wider implantation of constructed wetlands across the Lindisfarne National Nature Reserve (LNNR).

Timeframe	Action	Recommendation	Success Criteria	Key Challenge
Immediate (0-6 months)	Protocol Standardisation	Natural England to collect TON data (not just nitrate). Systematic sample procedure of predetermined parameters and frequency.	Standardised monitoring protocol established	Comparing data from different sources
	Kiln Point Construction	Proceed with 5-cell design, expected to return >96% reduction in TON. Undertake pre-construction baseline monitoring (1 month minimum)	Baseline data collected, construction commenced.	Higher input of TON than at Cockle Ridge. High variability in recent samples.
Short-term (6-18 months)	Cockle Ridge Optimisation	Formalise overspill pond as permanent 4th cell. Conduct summer maintenance when water levels low and focus maintenance on Wetland A	High treatment performance of TON sustained.	Ammonium concentration increasing at outflow.
	Wider LNNR catchment	Identify other TON-dominated catchments where CWs could be implemented. Consider single-cell systems for space-limited areas	Expansion plan developed	Funding opportunities
Long-term (18+ months)	Adaptive Management	Adjust maintenance frequency based on suspended sediment monitoring. Monitor ammonium trends (oligotrophication prevention). Regular vegetation composition assessment	Optimised monitoring and maintenance schedule	Maintenance capacity

For future CWs, implemented across the LNNR catchment and beyond, a series of five design recommendations have been summarised in *Table 7*, based on this study's findings. The primary recommendation is implementing at least one wetland cell in TON-dominated catchments as this study found that 77.5% of total TON reduction occurred in the first cell alone, making even a small intervention substantially better than none.

Table 7: Design and implementation recommendations for future constructed wetland interventions, including the rationale for each recommendation and any associated maintenance requirements.

Design Component	Specification	Rationale	Maintenance
Number of Cells	Three to four cells optimal (diminishing returns) or single-cell systems where space and budget constraints are limiting	77.5% treatment in first cell, marginal gains after cell 3	Focus on maintaining Cell 1
Cell Shape	Cells with larger surface areas and elongated from inflow to outflow.	Provide greater hydraulic retention time.	Regular maintenance to prevent sedimentation impacting cell size.
Vegetation	Prioritise emergent species over submerged	Higher nitrogen attenuation rates	Seasonal assessment and vegetation harvesting
Maintenance timing	Summer maintenance when low water levels	Easier with lower water depth and to minimise vegetation damage	Establish an annual summer maintenance window
Monitoring Infrastructure	Establish permanent sampling locations	Consistent data collection, reduces variability	Ensure same sampling locations are used by all agencies and researchers

5 Conclusion

This study demonstrates the effectiveness of CWs as NbSs for reducing nutrient pollution and mitigating estuarine eutrophication. Through assessment of past and present nutrient dynamics at the Cockle Ridge pilot intervention and the site of the proposed Kiln Point intervention, this research provides compelling evidence for the strategic implementation of CWs across the LNNR and other nitrogen-dominated catchments.

The Cockle Ridge pilot intervention achieved exceptional performance, with a 95% TON reduction compared to pre-intervention levels, and 96.4% reduction across the intervention itself. This performance substantially exceeds the UK government's 2038 target of a 40% reduction in nitrogen pollution and places the intervention at the upper end of reported CW effectiveness globally.

Rapid vegetation colonisation within nine months indicates successful wetland establishment, though algae development in Wetland A, while potentially contributing to nutrient removal through enhanced denitrification processes, highlights the need for monitoring of dissolved oxygen concentrations, and their impacts on vegetation and wetland processes.

The identification of TON as the dominant nutrient pollutant highlights the need for extensive nitrogen management for eutrophication prevention in Budle Bay. Analysis revealed that 77.5% of total TON reduction occurred in the first wetland cell, demonstrating that substantial water quality improvements can be achieved even with simplified single-cell interventions. This finding has profound implications for cost-effectiveness and scalability, particularly in space-constrained environments where multi-cell systems may not be feasible. The substantial TON reduction appears to be primarily driven by microbial denitrification rather than plant uptake or sediment adsorption, with the anoxic conditions created by algae mats probably enhancing this process.

The 88.3% increase in ammonium concentrations compared to pre-intervention levels, and 90.8% increase across the intervention, attributed to DNRA, require careful monitoring to identify potential oligotrophication impacts. However, with concentrations 99.4% lower than current TON input levels, ammonium does not cause environmental impacts at present.
The modest 11.6% phosphorus reduction compared to pre-intervention levels, despite a 66.9% reduction across the intervention itself, indicates that phosphorus management may require additional strategies beyond CW implementation. The plateau in phosphorus

removal performance in later wetland cells suggests that treatment capacity becomes limited, requiring maintenance to enhance effectiveness.

The Cockle Ridge pilot intervention represents a highly successful nature-based solution for nitrogen pollution management. The exceptional TON removal performance, combined with rapid vegetation establishment and clear understanding of treatment processes, provides compelling evidence for scaling this approach across similar environments. However, the identification of ammonium increases emphasises that CW implementation requires careful long-term monitoring and ongoing maintenance.

Success at Cockle Ridge provides strong justification for implementing CWs at Kiln Point, where higher TON concentrations suggest an even greater eutrophication risk. Extrapolated results indicate the proposed five-cell intervention could achieve a 96.59% reduction in TON. While the higher baseline nutrient concentrations at Kiln Point may require enhanced design specifications, including more frequent maintenance, to achieve optimal performance, further monitoring should be conducted to better understand nutrient dynamics at Kiln Point.

CW scalability and potential to address eutrophication challenges extends beyond the LNNR, such as Lough Neagh, where strategic CW implementation across tributary networks could significantly improve it's current 'poor ecological status'. Priority should be given to implementing at least one wetland cell in TON-dominated catchments, with three to four cells representing optimal cost-effectiveness given diminishing returns beyond this.

Critical future research priorities include a seasonal performance assessment, as this spring/summer study provides only partial understanding of CW’s year-round effectiveness. Establishing a coordinated, standardised monitoring methodology across agencies is essential, incorporating monthly sampling of dissolved oxygen, suspended sediment, temperature, pH, and all three nutrients to understand long-term performance and wetland maturation impacts.

This study demonstrates that properly designed and maintained CWs can provide effective solutions for estuarine water quality degradation, making substantial contributions to national water quality targets while protecting valuable coastal ecosystems.

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7 Appendices

7.1 Appendix I- Vegetation species observations at Cockle Ridge



Photo 1: CR Wetland



Photo 2: CR Wetland



Photo 3: CR Wetland



Photo 4: CR Wetland



Photo 5: CR Wetland A/B



Photo 6: CR Wetland B



Photo 7: CR Wetland B



Photo 8: CR Right side of Wetland C



Photo 9: CR Left side of Wetland C



Photo 10: CR Wetland C



Photo 11: CR Wetland C



Photo 12: CR Overspill/ Wetland C



Photo 17: CR Downstream



Photo 18: CR Downstream

7.2 Appendix II: pH data

site_id	wetland	cell_number	replicate	id	sample_type	wetland_avg_ton	wetland_avg_p	wetland_avg_nh4	ph
KP	proposed	na	1	KPSS1	sed	3.004838757	3.599310338	20.07454922	7.14
KP	proposed	na	2	KPSS2	sed	1.501225465	1.625257833	13.75632151	7.12
KP	proposed	na	3	KPSS3	sed	2.711989526	4.120109519	24.46491943	7.38
KP	proposed	na	1	KPSW1	water	1.73299	0.0004	0.056	7.17
KP	proposed	na	2	KPSW2	water	12.44194	-0.0046	0.049	7.26
KP	proposed	na	3	KPSW3	water	6.78874	-0.0006	0.023	7.06
KP	downstrea	na	1	KPDW1	water	9.02678	0.0064	0.027	7.23
KP	downstrea	na	2	KPDW2	water	4.63267	0.0014	0.016	7.22
KP	downstrea	na	3	KPDW3	water	8.42436	-0.0016	0.041	7.23
KP	downstrea	na	4	KPDW4	water	5.3312	-0.0036	0.024	7.47
CR	A	1	1	CRAS1	sed	0.169850427	107.5942254	103.2101707	6.31
CR	A	1	2	CRAS2	sed	0.243376166	79.21415277	91.76359091	6.39
CR	A	1	3	CRAS3	sed	-0.093467222	78.45338044	71.0983533	6.52
CR	B	2	1	CRBS1	sed	0.660662149	96.51467715	160.8091261	7.07
CR	B	2	2	CRBS2	sed	0.498977286	72.88850386	117.9402565	6.9
CR	B	2	3	CRBS3	sed	0.505502973	128.2225751	114.215766	6.54
CR	C	3	1	CRCS1	sed	0.132495156	155.7039067	166.9035455	6.63
CR	C	3	2	CRCS2	sed	-0.039326749	33.06717238	62.07021936	6.84
CR	C	3	3	CRCS3	sed	0.616293797	40.07956334	162.5870798	6.92
CR	downstrea	na	1	CRDS1	sed	0.305175347	32.86707026	213.4313339	7.15
CR	downstrea	na	2	CRDS2	sed	1.565889447	59.11573578	47.37402895	7.26
CR	downstrea	na	3	CRDS3	sed	0.48724327	76.67197451	49.22380036	7.22
CR	RS	na	1	CRRSS1	sed	33.05740546	178.7927491	70.22834121	7.35
CR	RS	na	2	CRRSS2	sed	30.84983937	106.8034854	71.05315321	7.13
CR	LS	na	1	CRLSS1	sed	4.363136805	88.17884483	65.02679857	7.09
CR	LS	na	2	CRLSS2	sed	9.088677481	69.694514	62.11472644	6.47
CR	upstream	0	1	CRUW1	water	3.32416	0.1344	0.025	8.59
CR	upstream	0	2	CRUW2	water	3.4321	0.1579	0.025	8.21
CR	upstream	0	3	CRUW3	water	6.15748	0.2189	0.03	9.09
CR	upstream	0	4	CRUW4	water	4.36261	0.2114	0.024	8.46
CR	upstream	0	5	CRUW5	water	4.01338	0.1504	0.023	7.86
CR	A	1	1	CRAW1	water	1.04678	0.1459	0.055	8.31

site_id	wetland	cell_number	replicate	id	sample_type	wetland_avg_ton	wetland_avg_p	wetland_avg_nh4	ph
CR	A	1	2	CRAW2	water	1.03712	0.0754	0.0475	7.76
CR	A	1	3	CRAW3	water	1.78192	0.0834	0.029	8.51
CR	A	1	4	CRAW4	water	0.30898	0.1324	0.046	8.31
CR	A	1	5	CRAW5	water	0.6909	0.0824	0.0515	8.37
CR	A	1	6	CRAW6	water	0.25151	0.0754	0.107	8.54
CR	B	2	1	CRBW1	water	0.98742	na	0.115	7.83
CR	B	2	2	CRBW2	water	0.74396	0.0584	0.1245	7.66
CR	B	2	3	CRBW3	water	1.39916	0.0769	0.193	8.27
CR	B	2	4	CRBW4	water	0.51002	0.0714	0.128	7.56
CR	B	2	5	CRBW5	water	0.82432	0.0829	0.19	7.79
CR	B	2	6	CRBW6	water	2.12758	0.0649	0.167	7.7
CR	C	3	1	CRCW1	water	0.182	0.0314	0.259	7.51
CR	C	3	2	CRCW2	water	0.08288	0.0504	0.286	8.26
CR	C	3	3	CRCW3	water	0.77532	0.0649	0.289	7.91
CR	C	3	4	CRCW4	water	0.20132	0.0804	0.506	7.77
CR	C	3	5	CRCW5	water	0.2044	0.0629	0.42	7.77
CR	C	3	6	CRCW6	water	0.10409	0.0704	0.406	8.39
CR	overspill	4	1	CROW1	water	0.18088	0.0554	0.179	7.48
CR	overspill	4	2	CROW2	water	0.39648	0.0514	0.1	7.56
CR	overspill	4	3	CROW3	water	0.20048	0.0749	0.21	7.66
CR	overspill	4	4	CROW4	water	0.11396	0.0504	0.401	7.37
CR	overspill	4	5	CROW5	water	0.11704	0.0474	0.432	7.93
CR	overspill	4	6	CROW6	water	0.06461	0.0709	0.2435	7.95
CR	downstrea	na	1	CRDW1	water	0.28588	0.0594	0.157	7.57
CR	downstrea	na	2	CRDW2	water	0.25102	0.0549	0.2675	7.65
CR	downstrea	na	3	CRDW3	water	0.17416	0.0594	0.23	7.63
CR	downstrea	na	4	CRDW4	water	0.16828	0.1124	0.306	8.3
CR	downstrea	na	5	CRDW5	water	0.33593	0.0564	0.317	7.65